國立臺灣科技大學

工業管理系

碩士學位論文

學號：M10901863

模擬機器人自動化系統倉庫揀貨訂單分配與根據相似性的組訂單批次處理之研究

Pick Order Assignment and Similarity-Based Group Order Batching using Simulation Approach for Improving Order Picking Performance in Robotic Mobile Fulfilment System Warehouse

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ABSTRACT

This study focused on order-picking system optimization by optimizing the joint performance problem between order batching and the pick order assignment in order-picking activities using a simulation approach. Order batching is used to classify the orders into some groups based on their order similarity. It will allow more orders processed concurrently to have a high affinity, allowing more SKUs to be picked from one pod and fewer pods needed. This study will also optimize the order-to-station assignment by considering the association between the new order and the currently active order being processed in the station instead of randomly assigning it. The objective is to provide a combined solution to deal with a dynamic environment and maximize the total throughput by increasing the pile-on value and minimizing the number of pods needed to fulfill the whole order. A simulation-based technique is used for solving the problem with two different scenarios: random as a baseline scenario and similarity-based order grouping and assignment as the second scenario.

The result shows that the proposed model significantly lowers the number of pods needed to fulfill the order. The proposed model outperforms and is competitive enough compared with the baseline scenario. Using similarity-based order batching and the order assignment model can decrease the average number of pods needed by 40%. It is in line with the pod's utilization percentage, which means dividing the number of pods needed by the total number of pods in the warehouse. Furthermore, putting an order with the highest similarity into the same batch has proven that it can increase the pile-on value of the system since it allows the simulation to achieve almost 70% better pile-on compared with the baseline scenario. The proposed model also sequences the order assignment to the station based on the similarity, allowing a higher chance of similar orders being processed concurrently. This strategy helped the system achieve higher throughput of around 3% by assigning similar orders concurrently within the batch and considering the active order in the picking station before assigning the batch. This order assignment strategy increases the possibility and ensures that every order and/or batch of orders always goes to the picking station with the most similar order. Then, it will help to maximize the pile-on, minimize the pod movement, and decrease the service time for each order fulfillment.

Keywords: Order Picking, RMFS, Pile-on, Pick Order Assignment, Order Batching, Simulation

ACKNOWLEDGEMENT

First, the greatest thank Allah for giving me mercy and grace so I can finish my study in Industrial Management National Taiwan University of Science and Technology. Second, I would like to express my sincere gratitude and appreciation to my advisor, Professor Shuo-Yan Chou, for always giving me suggestions, support, advice, and many opportunities. Then, I also want to send my gratitude to Professor Po-Hsun Kuo as my co-advisor and Professor Kung-Jeng Wang as my oral defense committee for their evaluation, suggestion, encouragement, and insightful feedback on my research. Finally, I am also grateful to my supervisor Dr. Anindhita Dewabharata and Dr. Ferani Eva Zulvia, for their continuous comments and suggestion throughout the study.

I would also like to thank all of my lab mates in the Information Technology and application Integration (ITAI) laboratory for motivating, helping, and being very supportive friends during my whole study. Thank you for all the time and beautiful experience I have never had before. Also, my biggest and warmest thanks to my best friends during my master's degree study Ko Edwin, Chrestella, Anggun, Elizabeth, and Rasyid. Very big thanks also to all of my friends in Indonesia and Taiwan who have been supporting me. Last, I would like to express my sincere gratitude to my family, especially my Mom and Dad, for always loving and supporting me.

Aisyahna Nurul Mauliddina

Taipei, January 2023

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CHAPTER 1
INTRODUCTION

1.1 Background

During the recent decade, the online shopping trend has overgrown year by year, particularly during the pandemic. Statistical data shows that e-commerce sales worldwide are expected to reach 21.8% of all retail sales worldwide by the end of 2024 [1]. The rapid growths of e-commerce and ‘lot size one’ production demands, driven by changing customer expectations of what ‘on demand’ means, have forced companies to evolve into more efficient, agile, and full-time operational epicenters. On the other hand, the industry, particularly the manufacturing sector currently facing significant changes in response to customer demand for increasingly diversified goods with high-quality requirements. They must compete with other companies by providing fast order fulfillment services. Companies leverage Industry 4.0 disruptive technologies to establish smart manufacturing and warehouses and renew their supply chain network [2]. In line with the growth of e-commerce and customer demand, production and warehouse operations are also increasing in complexity with time [3].

Moreover, the performance of the customer’s order fulfillment time is affected by the supply chain and logistics activities. It becomes a new challenge for warehousing operations since e-commerce involves tremendous small orders, facing uncertain and enormous customer demands such as rapidly changing trends and promotional activities. Online retailers need to fulfill orders as soon as possible. To fulfill huge orders, the warehouse has a critical role in the whole supply chain since all of the warehouse’s internal operations need to be done effectively and efficiently.

The warehousing operations activity consists of several divisions, from receiving, storing the product, order picking, replenishing, and shipping the order. The order-picking activities contribute about 55% of the total cost of warehousing activity [4]. Order picking also faces challenges such as time, accuracy, technology, and costs. It depends on which type of order-picking methods that applied in the warehouse. The order-picking techniques can be divided into manual and automated order-picking. The old warehouse utilizes manual order picking called picker-to-parts. Picker-to-parts method is no longer relevant if faced with effectiveness and efficiency issues. Currently, many warehouses have already employed and developed the parts-to-picker method. The use and development of the parts-to-picker way are increasing since companies have struggled to source enough skilled workers to keep pace with the size of ramped-up operations. The labor availability, the increasing volume and throughput of work, and the need to address new business requirements are three main drivers that make companies need to change their technology [5]. As new automation systems enter the market, technology plays an essential role. Many companies are trying to invest in automation and robotic technologies that increase their company's competitiveness [6].

A company needs to analyze and understand which technologies they must apply and what kind of impact they can take from the new technology. The right level of automation will not only enable faster, safer, and more efficient warehouse operations [7]. It will also reduce costs and improve delivery times, resulting in a quicker, leaner, more scalable, and more sustainable process that will eventually provide the end-consumer with the high service level needed in the modern on-demand economy [2]. The use of automation can be implemented either partially or fully in warehouse activity. As previously said, order picking activity contributes the majority of warehousing costs. Therefore, automation in this activity is critical to lowering order-picking costs while increasing the picking process's effectiveness, efficiency, and accuracy. Automated material handling systems tend to have much lower error rates than manual order picking, improve the accuracy of inventory management and tracking, and can achieve higher pick rates. One of the implementations of automation in the order-picking process is using a Robotic Mobile Fulfillment System (RMFS).

RMFS is an automated parts-to-picker system that uses an autonomous robot to carry the movable racks (pod) from the storage to the picking/replenishment station or vice versa [8]. RMFS uses Automated Guided Vehicle (AGV) to bring pods containing ordered items to the picking station to fulfill an order, bring the pods back to the storage for storing the pods, or bring them to the replenishment station for replenishing the SKU inside a pod. The RMFS has developed in many well-known companies such as Amazon, Geek+, and Fetch Robotics. The use of RMFS is promising because this system can help to increase the picking rates since RMFS will reduce the picker walking time compared with the picker-to-parts system. Implementing RMFS also improves the order fulfillment speed by 50% faster than traditional warehouses [9].

On the other hand, several issues must be addressed when a company wants to implement RMFS. The problems related to optimizing their layout and facility, warehousing activities such as order assignment, product storage assignment, pod storage allocation, and robot routing problem. The total throughput usually measures the efficiency of the warehouse. The total throughput means the total completed orders per unit of time. Therefore, optimization from all warehousing processes is needed to achieve higher throughput.

Several decision problems in the RMFS warehouse need to be considered, including how to optimize the order-picking process in the warehouse to maximize the total throughput. Since the system's performance follows the determined decision rules, the unit throughput rate in this system is greatly affected by the decision rules, including Pick Order Assignment (POA). POA is the activities to fulfill all orders in the warehouse, including how the system will assign the order to a pod and a particular picking station. Fulfilling more orders with a lesser number of robots or a lesser number of pods to be picked is one of the crucial issues for optimizing the order-picking process in RMFS. In this kind of warehouse, one robot only can carry one pod every single tour, but not all of the items in the order can be fulfilled by one pod. The system needs an effective and efficient assignment model so more items can be picked from one pod. Pile on value is a performance measure representing the average number of items picked from one pod. A higher pile on value means more items can be picked. Fewer pods are required to be delivered to the station. Order batching is used to process all orders effectively, where several orders are grouped in the same pick order to reduce the traveling time and order completion time.

The order batching rule assigns several orders in one pick order and the same picking station, so it might reduce the total order completion time by taking advantage of fulfilling many orders simultaneously. The main objective of order batching is maximizing total throughput. It can be achieved because applying order batching helps reduce the average completion time by minimizing picking and travel time needed. In the concept of RMFS, order batching is an efficient way for order assignment, increasing the pile-on value and minimizing the number of pods needed to be delivered to the picking station. The application of order batching might follow specific rules. The use of order batching might help to increase the performance of picking activities. However, the performance of the order picking still depends on the order assignment rule on how the system will execute the batch order in terms of assigning it to the picking station and pods. A good assignment will optimize the system, significantly affecting the total warehouse throughput.

Developing a mathematical model is sometimes constrained by the model's ability to capture the real warehouse problem and dynamic factors. In this situation, simulation is needed to test the model's reliability for solving the problem most similar to the real problem. The development of technology and digitalization allows a company to utilize digital warehouse design by creating a "digital twin" of their current warehouse facility. This digital twin simulation can be used to model the impact of changes to the layout and workflow before moving physical assets, making investments, or changing the warehouse [3]. The use of simulation certainly provides an excellent opportunity for companies to determine the configuration of facilities and warehouse layouts as well as operational activities in the warehouse in line with the optimal results from the simulation. The improved warehouse design resulting from the simulation can help companies improve their efficiency by 20-25 percent before spending money to test and execute the improvements [3]. The simulation can visually represent the actual system and allow us to compare several scenarios easily, effectively, and efficiently.

Several previous researches tried to propose mathematical models for optimizing both the order batching problem and the order picking assignment problem. Unlike previous research, this research will use a simulation system platform to implement the proposed model for optimizing the pick order assignment by considering the advantages of the orders batching problem. It allows this research to simultaneously solve the order allocation and order assignment problems. In this research, the batch size will change dynamically depending on each picking station's available capacity. Furthermore, similarity-based order batching will allocate the most similar orders in the same pick order list to increase the possibility of achieving a higher pile-on and minimizing the number of needed pods. The similarity is also used to sequence the order inside a batch in the order assignment rule, allowing orders with higher similarity to be processed concurrently and minimizing the pod movement.

1.2 Objectives

This study will focus on the order-picking process in the RMFS warehouse called Pick Order Assignment, shortened as POA. This research aims to increase the total throughput by maximizing the pile-on and minimizing the total number of pods needed for fulfilling the order. The pile-on number represents a pod's levels of utility since a high pile-on value indicates that a pod can fill more SKUs in a single trip. Because the more excellent the pile-on value means, the more likely the order list may be fulfilled faster. Generally, the pile-on value is also related to the number of pods required to satisfy orders. This condition happened because the larger the pile-on number, the fewer pods will likely be needed to finish the order list. As a result, there were fewer pods transported to the picking station. Then the AGVs did not have to move as much throughout the warehouse. Besides that, fewer pods delivered must optimize pod use to ensure warehouse throughput efficiency.

In this research, order batching will be applied to the order assignment, which means the order will be grouped into a batch based on specific rules when certain orders have arrived. Finally, the proposed research tries to optimize the system using a simulation approach by allocating the order inside the same batch, considering the similarity factor and how to simultaneously assign the batch order to the picking station and pod. The detailed research framework and scenarios will be explained in Chapter 3.

1.3 Scope and Limitations

The following is a comprehensive overview of the scope and limitations of the present research:

- All robots are allocated to both pick-up and replenishment stations, then the number of AGV will not be differentiated
- The study will primarily examine POA, leaving out other warehouse-related tasks
- The replenishment process was done entirely before the order-picking activity
- The inventory level of all SKUs is enough to fulfill all orders on that day
- Each order consists of several SKUs with a specific quantity
- The system is simulated using NetLogo as the primary program and platform for Agent-Based Modelling. This simulation software is integrated with Python as the optimization platform and Microsoft Excel as the database platform to store simulation data and results.

1.4 Organizations of Thesis

This research has been written and divided into five chapters listed below. The first chapter describes the research's background and purpose, as well as its aim, scope, and limitations. Chapter 2 reviews the literature on related subjects and highlights previous research. The third chapter discusses in greater detail the research methodology, including the research framework, problem formulation, and performance indicators utilized in the research. The data processing and experimental results will be explained and presented in Chapter 4. The conclusion and suggestions for further research will be listed in Chapter 5.

CHAPTER 2
LITERATURE REVIEW

2.1 Order Picking in RMFS Warehouse

Order picking deals with retrieving items to satisfy a given demand, which plays the most vital role in operating warehouses. Order picking is the process of retrieving products from the warehouse inventory to fulfill customer orders that involve the process of clustering and scheduling customer orders, releasing them to the floor, picking items from storage locations, and disposing of the picked items[10]. Order picking itself can be divided into manual handling and automated handling. The detailed classification of the order-picking system can be seen in Figure 2.1.

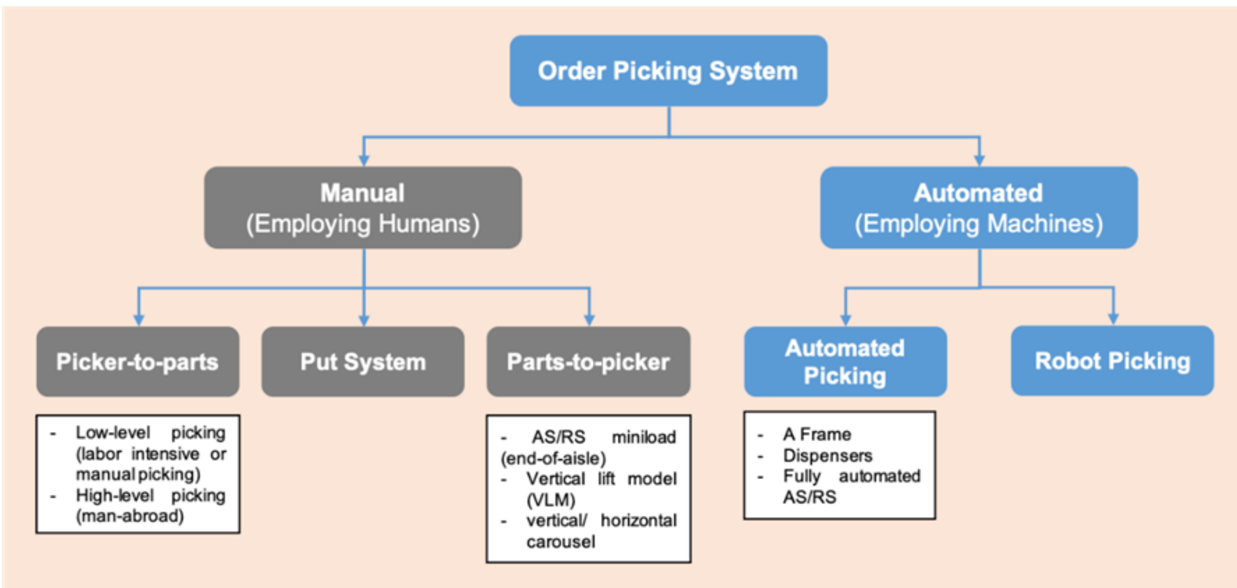


Figure 2.1 Order Picking System Classification[10]

For manual handling, the picker-to-parts system is the oldest, also called traditional order picking, the most commonly used warehouse picking method. This system allows the picker to go to the storage location and takes the required quantity of goods stored there using hand or simple material handling. Even though this method only needs low procurement, installment, and maintenance costs, it has many disadvantages, including the high chance for human error, time consumed, low picking accuracy, and not having a real-time system. Many warehouses already use parts-to-pickers as their primary order-picking method, especially for high turnover with many products warehouse. Parts-to-picker can reduce the labor cost since it should need fewer human workers. This system also gives higher picking performance, increased productivity and efficiency, better warehouse floor space utilization, and better workplace safety conditions [11]. Nevertheless, applying this system needs high procurement and installment costs in the beginning, high maintenance and service costs, and the improvement of technology in the overall warehouse system.

2.2 Robotic Mobile Fulfillment System

A robotic mobile fulfillment system is an automated parts-to-picker material handling system designed to fulfill e-commerce orders effectively. The system consists of pods, AGVs, and picking and replenishment stations. The representation of the elements in RMFS can be seen in Error! Reference source not found.

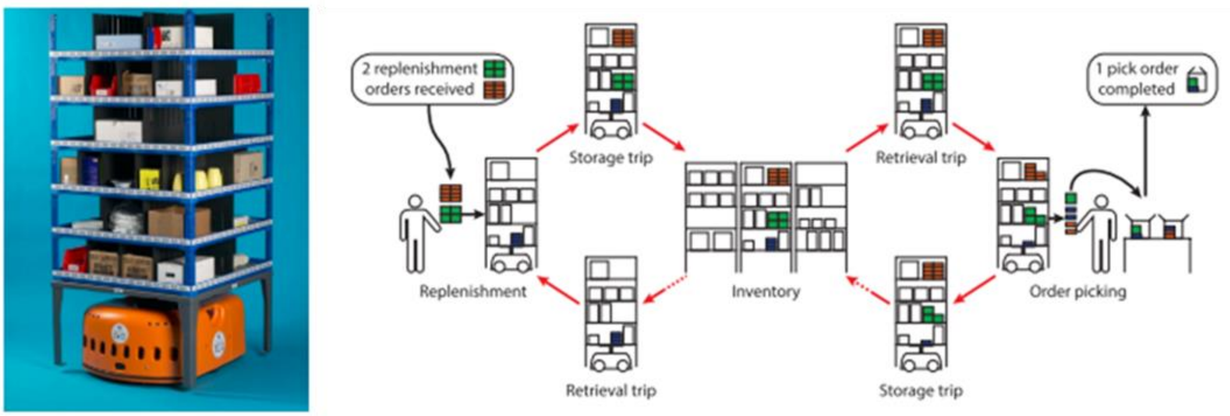


Figure 2.2 Elements of Robotic Mobile Fulfillment System [12]

The process of RMFS starts when orders arrive in the system. After particular order arrives, the system will assign each order to a specific picking station. Every picking station can fulfill several orders simultaneously, depending on the available capacity in each picking station. If an order arrives when there is no available capacity in the station, that order will be added to the order pool containing unassigned orders. On the other hand, every pod stored in the warehouse can include multiple items depending on the item assignment and replenishment policy. Each pod only can serve or go to one station at a time. So, the system will search for possible pods containing the specific item for fulfilling current active orders in the picking station[13]. The overview of the RMFS warehouse layout operation can be seen in Figure 2.3.

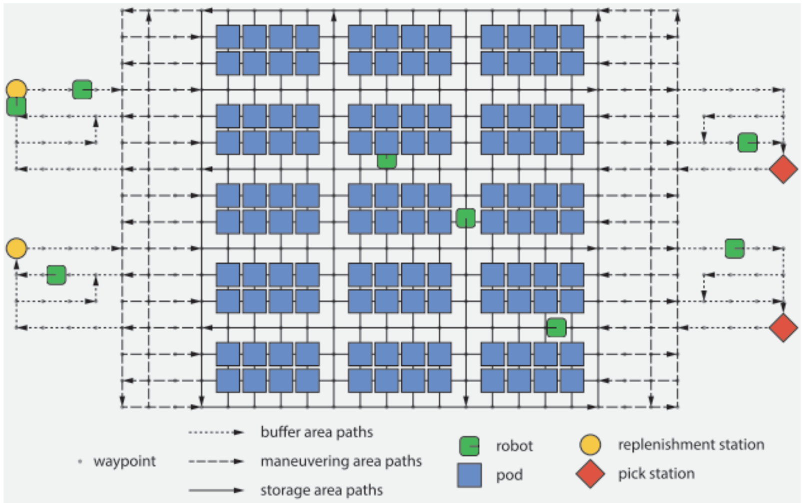


Figure 2.3 RMFS Layout and Warehouse Operation[11]

The Orders assignment and pods selection problem are crucial elements affecting picking efficiency [13]. There are some decision problems in an RMFS warehouse that can be divided into three levels include strategic level, tactical level, and operational level. The hierarchical overview of decision problems in RMFS can be seen in Figure 2.4. Operational decision consists of several decision problems below.

1. Order Assignment (order to station assignment)
    - POA, assignment of pick order to the picking station
    - Replenishment Order Assignment (RPA), assignment of replenishment order to the replenishment station
2. Task Creation (for pods)
    - Pick Pod Selection (PPS), pods selection for picking activity
    - Replenishment Pod Selection (RPS), pods selection for replenishment activity
3. Task Allocation (TA) (for AGVs) generates the task sequence for AGVs in order to fulfill the order
4. Path Planning (PP) (for AGVs), path generation for AGVs to follow.

The decisions made to address certain operational issues can affect one another. Two specific relationships between decision problems can either work together to create positive outcomes or undermine each other's success—for example, the relation between POA and PPS. One goal is to increase the pile-on value. A higher pile-on value means more SKUs can be picked from a pod every time they visit the picking station. In other words, the higher the pile-on, the fewer pods are needed at pick stations. Therefore, the choice of a pod for an order in PPS depends on the pick station chosen for other orders. If similar orders are assigned to the same pick station, a pod can be selected to maximize the number of items picked. This research will mainly focus on the operational level, specifically on the POA part.

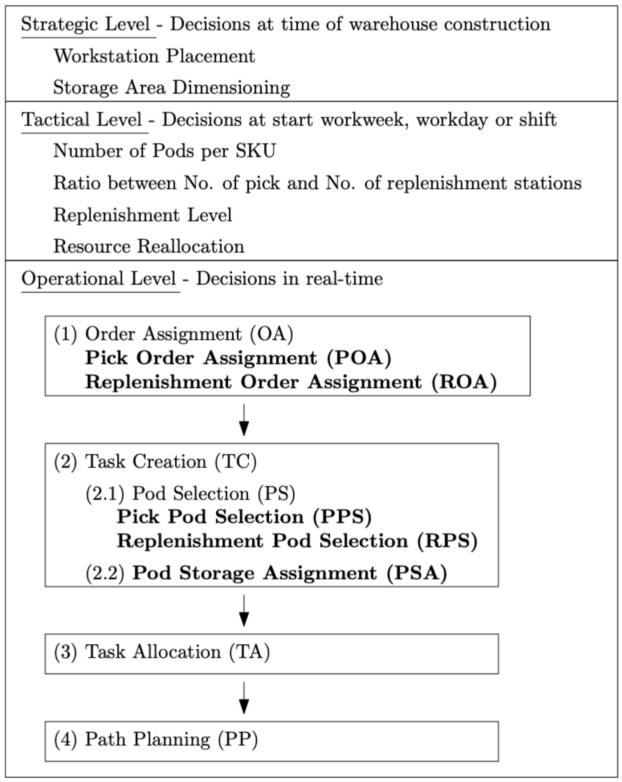


Figure 2.4 Decision Problem in Robotic Mobile Fulfillment System [12]

2.2.1 Pick Order Assignment (POA)

In a warehouse or distribution center, allocating a new pick order to a picking station after a previous order has been completed is referred to as POA or pick order assignment. This process is essential for ensuring that orders are completed promptly and efficiently. One key metric for measuring the efficency of POA is pile-on, which is defined as the average number of SKUs collected from a storage pod during each visit to a picking station [13]. Pile-on is an essential indicator of how effectively orders are assigned to pick stations. It can impact the overall time it takes to complete an order and the number of required resources (such as pods and AGVs). Generally, a higher pile-on is preferable as it indicates that pickers can collect more units from each pod during their visits to a picking station, reducing the number of times pods need to be moved between picking stations and storage areas.

When an order arrives, a picking station must be selected and submitted to the system. Ideally, the new order will arrive as soon as an existing order is removed from the order pool so that the picking station can be quickly reassigned to the following order. It is known as the "fast replacement" of orders at the station. By ensuring that orders are quickly and efficiently replaced, this process can increase the availability of picking stations and reduce the time it takes for orders to be completed. Numerous rules can be used to determine which pick order should be assigned to a specific pick station [14]. These rules can take into account factors such as the order's priority, the amounts of units it contains, and the current availability of pick stations. The specific rules used will depend on the warehouse's layout and operations. However, regardless of the approach, the goal is to assign orders to pick stations to maximize efficiency and minimize the time it takes for orders to be completed.

Overall, POA is a critical process that plays a key role in a warehouse or distribution center's efficiency and performance. By carefully managing the assignment of orders to pick stations and continually monitoring performance metrics such as pile-on, warehouse managers and operators can ensure that orders are completed quickly and effectively, with minimal use of resources. Also, numerous rules exist for picking the following order from the order pool to fill an open slot capacity at a picking station, as discussed below.

1. Random: This rule randomly assigns the following order from the orders pool.
2. First Come First Serve (FCFS): Assigning orders that are received first to the picking station.
3. Due Time: This method will assign orders with the earliest due date from the order pool but also maintains the goal of completing the orders before the deadline.
4. Fast-Lane: Fast-Lane assigns the order in a similar way similar to the Random rule. However, this method will reserve one slot capacity to complete specific orders quickly.
5. Common Lines: This rule compares the pick stations with the orders already assigned to them to all orders in the pool and assigns the order with the most common lines
6. Pod-Match: Pod-Match assigns the order to depend on the most matched order with pods that are moving to the picking station at that moment.

2.2.2 Order Batching in Order Picking Assignment

Order batching is a technique that groups customer orders into larger picking orders. This approach is often used to improve warehouse efficiency and reduce operating costs, as it can minimize the time and distance required for manual picking. Order batching aims to split large orders into smaller, more manageable groups optimized according to specific rules and objectives. These objectives may include minimizing make-span [15], reducing workload [16], and increasing the similarity of items within each batch [17]. It is a well-established field of research and an NP-hard problem that is hard to solve exactly. Importantly, it is typically a one-time assignment, meaning the grouping of orders is done only once and then picked.

Order batching can be classified into two categories: static and dynamic. In static order batching, all details of customer orders, including the products and quantities required, are known before the orders are grouped together. On the other hand, in dynamic order batching, the information about the orders becomes available only when the orders arrive. This means that orders may arrive at different times while the picking process is already in progress. The main challenge in dynamic order batching is identifying the optimal batch size and timing to minimize the time required to complete all orders. Unlike traditional order batching research, dynamic order batching involves selecting and processing orders together, considering the number of active products a workstation handles. However, the batches in the dynamic order batching setting are often subject to change due to the varying processing times of different orders. As a result, the batching process becomes an ongoing, dynamic sorting of orders based on their actual processing times.

2.3 System Simulation

System simulation is a process that involves creating a virtual model of a real-world system and then analyzing its behavior in a controlled environment. The process is typically divided into the following steps:

- Define the system: The first step in building a simulation is clearly defining the modeled system. It includes identifying the system’s boundaries, the inputs and outputs, and the relationships between the different components of the system. It is essential to have a clear understanding of the system's structure and behavior in order to model it accurately.
- Develop a model: The next step is developing a mathematical or computer-based system model. This model should be based on a set of assumptions about the system's behavior,

and it should be able to represent the key characteristics of the system accurately. Depending on the system's complexity, this model can be a simple mathematical equation or a complex computer program.

- Validate the model: Once it has been developed, it is crucial to validate it to ensure that it accurately represents the real-world system. This validation can be done by comparing the model's outputs to real-world data, such as measurements of a physical prototype, or by conducting experiments on the physical system itself. This step is crucial to ensure that the model can be trusted to provide reliable results.
- Conduct the simulation: Once the model has been validated, the simulation can be run to observe the system's behavior over time. This simulation can be done by inputting various scenarios or conditions into the model and observing the output. It is a way to test the model in a controlled environment, considering all kinds of scenarios.
- Analyze the results: After the simulation has been conducted, the results should be analyzed to understand the system's behavior and identify potential areas for improvement. This analytics can include identifying any unexpected or abnormal behavior and any points where the system's performance could be improved. The analysis results can be used to refine the model and improve the overall accuracy of the simulation.

Overall, system simulation is a powerful tool for understanding the behavior of complex systems and identifying ways to improve their performance. It allows us to test the system in a controlled environment, considering all possible scenarios, which is not always feasible in real-world scenarios.

2.4 Research Comparison

Previous research tried solving the order assignment problem separately or combined with other decisions. The detail can be seen in Table 2.1. A joint optimization model of order sequencing and rack scheduling has been proposed by [13]. They deal with synchronizing the sequence of pending orders and the sequence of arriving pods jointly. They have proven that an optimal order sequencing and pod sequencing policy can minimize the total number of robotic tasks. Based on their results, the scenario can maximize the pile-on so the total throughput will be higher, and the completion time also can be reduced.

Table 2.1 State of Comparison for Current Research

No	Title	Order		Pod		System	Methodology	Type of Order Batching
		Allocation	Sequencing	Allocation	Sequencing			
1	‘Minimizing Order Picking Make-span with Multiple Pickers in A Wave Picking Warehouse’[18]	✓	✓			Picker to parts	Metaheuristics	Static
2	‘Spatial and Temporal Optimization for Smart Warehouses with Fast Turnover’[15]	✓				Picker to parts	Column Generation	Static
3	‘Order Batching in A Pick-and-Pass Warehousing System with Group Genetic Algorithm’[16]	✓				Pick and pass	MIP, Group Genetic	Static
4	‘Order Sequence Optimization for ‘part-to-picker’ Order Picking System’		✓			AS/RS	MIP, Clustering	Static
5	‘Parts-to-picker Based Order Processing in A Rack Moving Mobile Robots Environment’[14]		✓		✓	Kiva System	MIP, DP-SA	Static
6	‘Efficient Order Processing in An Inverse Order Picking System’[19]		✓		✓	Put to light	MIP, Heuristic	Static
7	‘Optimization of put-to-light Order Picking System’[20]		✓			Put to light	Local Search	Static
8	‘Optimal Selection of Movable Shelves Under Cargo-to-person Picking Mode’[21]			✓		Cargo to person	IP, Heuristic	Static
9	‘Order Allocation, Rack Allocation, and Rack Sequencing for Pickers in A Mobile Rack Environment’[22]	✓		✓	✓	RMFS	MIP, CPLEX	Dynamic
10	‘Joint Optimization of Order Sequencing and Rack Scheduling in The Robotic Mobile Fulfillment System’[13]		✓	✓	✓	RMFS	MIP, Heuristic	Dynamic
This research	Pick Order Assignment Optimization and Similarity-Based Group Order Batching for Improving Order Picking Performance in Robotic Mobile Fulfilment System Warehouse	✓	✓	✓		RMFS	Heuristic, Simulation	Dynamic

The advantage can also be achieved since this scenario allows us to minimize the number of robotic tasks and pod movement, so there is an opportunity to reduce the AGVs needed and the waiting time. Also, it can minimize the total energy consumption for each AGV so we can minimize the material handling cost and total cost in the end. The limitation of this research is that they have not considered how to put the orders inside the batch.

In this research, the term of assigning order both to station and pod are tackled simultaneously. The order assignment to the picking station follows a heuristic rule based on the SKU's similarity between orders. In addition, pod assignment will follow the most pile-on, which means the pod that contains the most SKUs needed in the picking station will be selected. The order batching process is also being utilized to group orders based on their similarity into pick orders. The batch size dynamically changes depending on the available capacity in the picking station. The SKUs similarity measurement is used to convince that orders with a higher similarity will be grouped in the same batch if there is more than one available capacity in the picking station. Furthermore, the similarity is also used for sequencing the order to the picking station, so orders with a higher similarity will be processed consecutively to minimize the pod movement. This research considers the dynamic batching and SKU similarity between the order that has not been considered in most of the previous research.

CHAPTER 3
METHODOLOGY

3.1 Problem Definition and Research Framework

This research will present the order-picking problem in Robotic Mobile Fulfillment System (RMFS) warehouse. Order picking is an activity for fulfilling all of the customer's orders. In the RMFS warehouse, order picking activities using a robot AGV instead of a picker go to the storage area. The AGV will receive a task for lifting the pod in the storage area to go to the picking or replenishment station. Otherwise, after a pod finishes its task in the station, the AGV will return the pods to the storage area. The process flow for order picking in the RMFS warehouse consists of several activities, including order-to-station and order-to-pod. This study will assign the order to the station, meaning that one order can only be assigned to one designated picking station.

In the RMFS warehouse, the picking station has bin capacity. Each bin only can be used for one order. So, the number of orders assigned to one picking station cannot exceed the total bin capacity for each picking station. In this study, the system also will utilize an order batching system based on batching time. It means that at every specific batching time, the orders in the order pool will be released and assigned to the picking station. The batching rule will assign the order to several groups as much as how many picking stations are available to process the order. The number of batches generated cannot be more than the number of available picking stations since one batch only can be assigned to one picking station. In addition, each order also only can include in one specific batch. It allows no overlap for one order assigning in more than one batch and one picking station. In terms of how many orders will be put in one batch, it depends on the number of bins available for each picking station. It allows the system to dynamically change the number of batches and size of the batch every time and in every situation.

In other research, the grouping order into a batch always follows a random rule, which means no specific criteria are used for assigning two or more orders to the same group. This study will use similarity-based order batching to assign orders into a batch based on the similarity of SKUs. The Jaccard index measurement maximizes the order similarity inside a batch. This order similarity also will be used for assigning the order to a station using the heuristics rule. The batch will be assigned to a station with the highest similarity with the station’s currently active order. Every order assigned to the batch and picking station is the order that the current pods' inventory can possibly fulfill. If no pods can fulfill an order, the order will be kept in the order pool.

In terms of assigning order-to-pod, the most pile-on rule will be applied in this problem. The system will assign the pods with the highest number of SKUs that can be picked to fulfill the order in the picking station. In this case, one pod only can serve one picking station at a time. However, after finishing one station, the pods can be utilized by another picking station directly. This study will be conducted following the research framework shown in Figure 2.1.

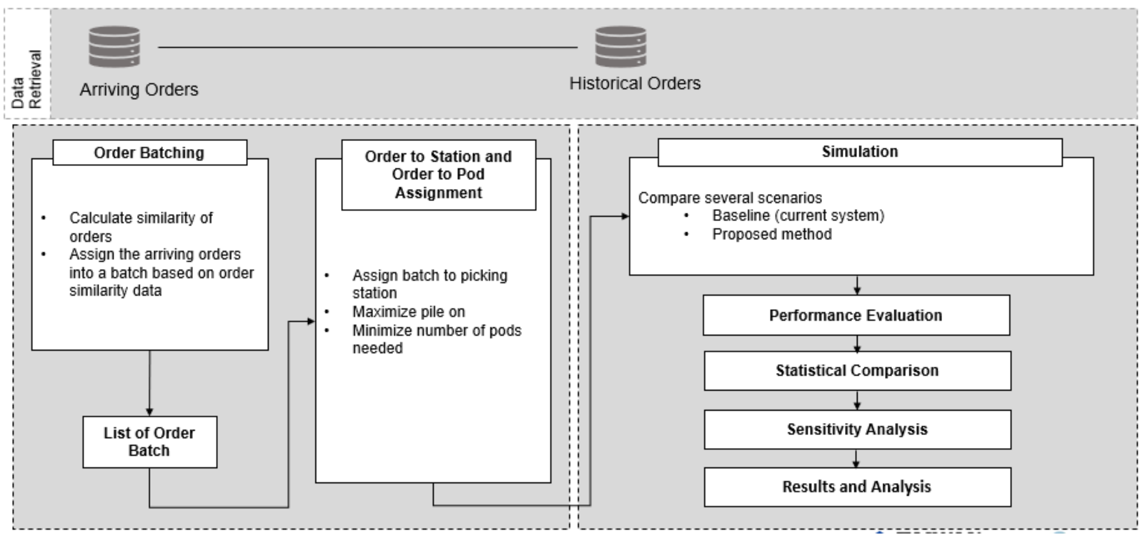


Figure 3.1 Research Framework

Using the similarity-based grouping rule, the proposed model will consider order batching and POA problems. The study will be divided into ordering batching and assignment to the picking station. However, both processes can be done simultaneously and related to the real system. Order batching in this system will use Jaccard Index as a measurement to examine the similarity between orders. After calculating the similarity, orders will be grouped based on their SKU's similarity to maximize the relationship between orders in the same group. The number of orders inside each batch will depend on the available capacity of each picking station, so the batch size is dynamically changing in real time. Then, the system will use similarity-based order batching rule to assign orders to the same batch. The total number of group batches will be the same or less than the number of picking stations, so each batch will only be assigned to one picking station.

Instead of only grouping the order based on their similarity, the similarity measurement was also used to assign which batch to send to which picking station. It means that the order assigned to a picking station also considers the similarity between the new order and the current processing order in the picking station. In terms of more than one capacity available in the picking station, the order will be sequenced using the heuristics method. The detail process of it is shown in Figure 3.2

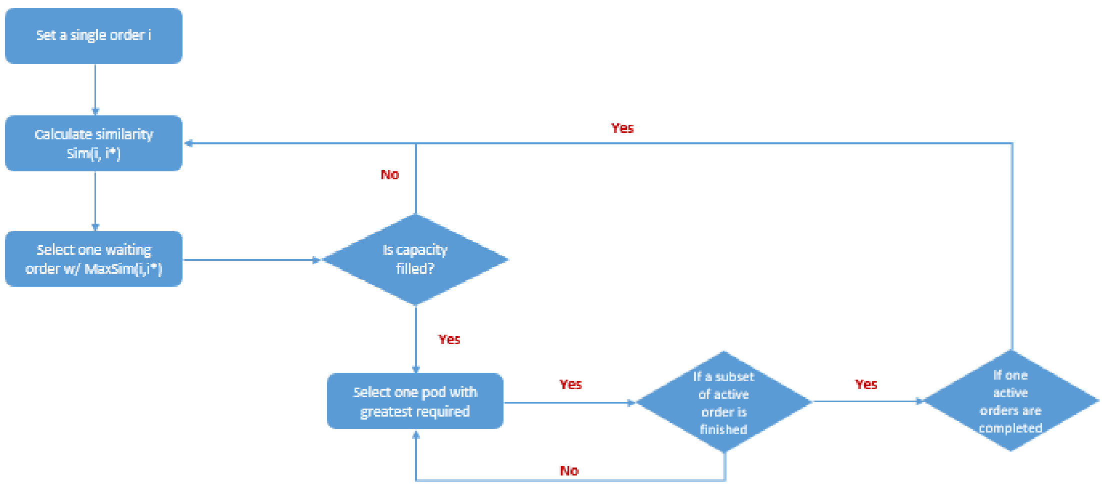


Figure 3.2 Order-to-station Sequencing Rule

3.2 System Process Flow

This research will focus only on pick order assignment, but the other system components and processes are still needed to simulate the results. The system process flows from order arrival until all orders are fulfilled will be described below.

1. Order Arrival

   The RMFS warehouse system started when amounts of orders arrived continuously.

2. SKU to Pod Assignment

   SKU-to-pod assignment means assigning and storing the SKUs inside different pods to maximize the pod's utility. This inventory strategy task involves dividing and classifying the SKUs based on the chosen rule. In this research, the inventory strategy will apply the ABC rule classification to store the SKUs in the warehouse. The proportion of the number of units is 60% class A, 25% class B, and 15% class C. The storage arrangement will follow a mixed-storage sharing policy so that each pod will have different SKUs and types.

3. Order to Pod Assignment

After orders have arrived in the system, the orders will be assigned to the picking station, and a list of pods that can fulfill the order is generated. For each pod, they might satisfy more than one order. In terms of assigning the order to the pod, the most pile-on rule is implemented in this system.

4. Pod Sequencing

After the system receives orders, it will assign the order to a specific picking station and generate a list of pods that can fulfill the orders. One pod can be assigned to fulfill more than one order. One or more pods can complete one order. Regarding several pods needing to fulfill one order, the sequence of the selected pods indicates the pod priorities to be picked to fulfill the order.

5. Robot to Pod Assignment

AGVs will be allocated to the chosen pods based on proximity and the earliest due date. Take twice as many available AGVs from the list of chosen pods, then assign AGVs to selected pods. However, the distance between the AGV and the chosen pod is computed using the Manhattan distance, and the Hungarian method is used to determine the assignment.

6. Robot Routing

AGV's routing policy employs a Simple Routing with a traffic control policy following prior research [47]. The traffic policy used is congestion and collision avoidance. This primary method in AGV routing allows it to have a minimal route without path planning to reduce the calculation time.

7. Robot to Workstation Assignment (Picking/Replenishment Station)

The assignment between the robot and the workstation is determined by the minimal number of AGVs waiting in line. It is related to how many AGVs need to be processed when the robot reaches the highway in the storage area before traveling to the picking station.

8. Robot to Storage Assignment

After a pod finishes its task in the order picking or replenishment station, it must be stored in the storage area. The system will put it in the nearest empty location available.

9. Pod to Replenish

After the pod finishes its task in a specific order-picking station, the system will check the pod inventory level. If the inventory level falls below the minimum inventory limit, it means the pod needs to be replenished. Then, the designated AGV will directly carry this pod to the replenishment station.

The overall RMFS process flow can be seen in Figure 3.3.

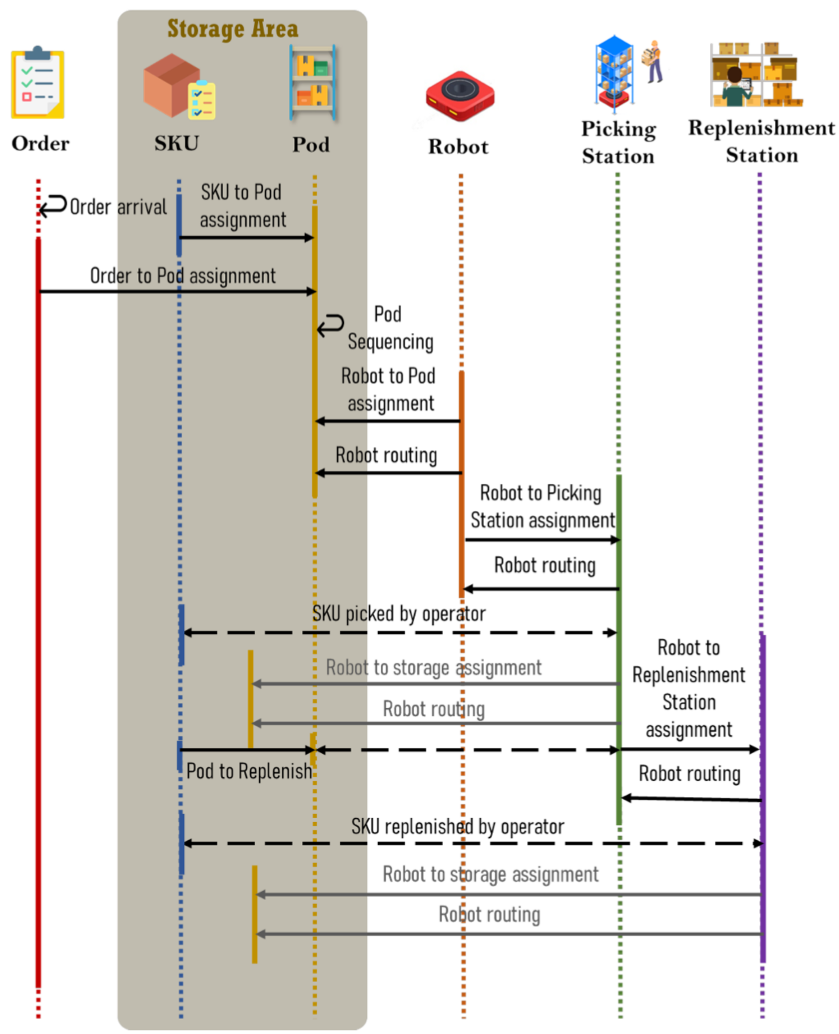


Figure 3.3 Robotic Mobile Fulfillment System Process Flow

3.3 System Configuration

This subchapter describes the system architecture, simulation layout, parameters, and other assumptions in the research.

3.3.1 System Architecture

In this research, the system is developed using NetLogo, Python, and Microsoft Excel as integrated platforms. The architecture of the system is shown in Figure 3.4. NetLogo is a simulation tool to visualize the warehouse operating process in RMFS. It is a modeling and simulation technique that allows agents to interact with one another. Agents may represent either physical elements, such as AGVs and workstations, or virtual ones, such as supervisors of the picking and restocking processes. Python is a programming language that manages system functions such as assignment outcomes. The system functions are native to NetLogo and interface with Python to provide optimization feedback. Microsoft Excel serves as the database and receives inputs and outputs from CSV files from NetLogo and Python, including layout design, orders, settings, pods, and items.

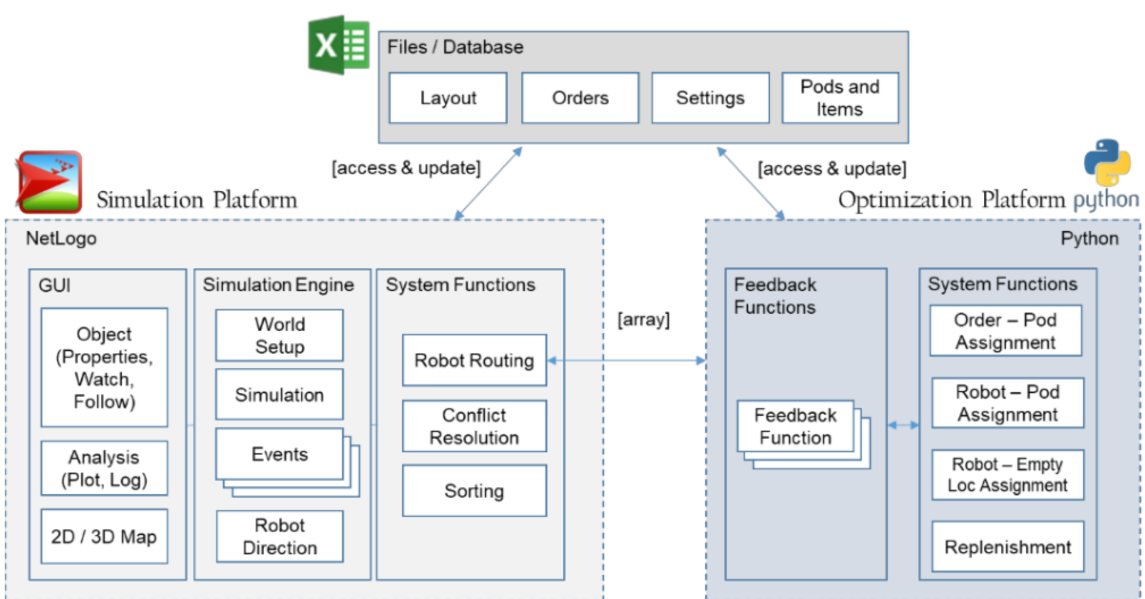


Figure 3.4 System Architechture

3.3.2 Simulation Layout

The simulation layout used in this research can be seen in Figure 3.5. The warehouse layout includes picking stations, a storage area, and replenishment stations. It depicts the NetLogo integrated warehouse layout arrangement. There are picking stations with pickers and queue lines at the top of the layout. There is a storage room holding pods carrying SKUs in the central part. A pod with a distinctive color indicates that it has been allocated to orders that AGV will pick in the future. The vacant storage space with the blank square inside the pods means that the AGV might return the pod to that spot once it has been selected. AGV can operate in a one-way aisle and go underneath the pod without direction. Within the aisle, the AGV can only move forward or stop. When it reaches a junction, it might turn toward the next aisle. Near the workstations, two-lane highways provide many routes to and from the workstations. According to their roles, there are two kinds of AGVs, which may be categorized as picking AGVs and replenishment AGVs. At the bottom of the layout, it also has replenishment stations and queue tracks.

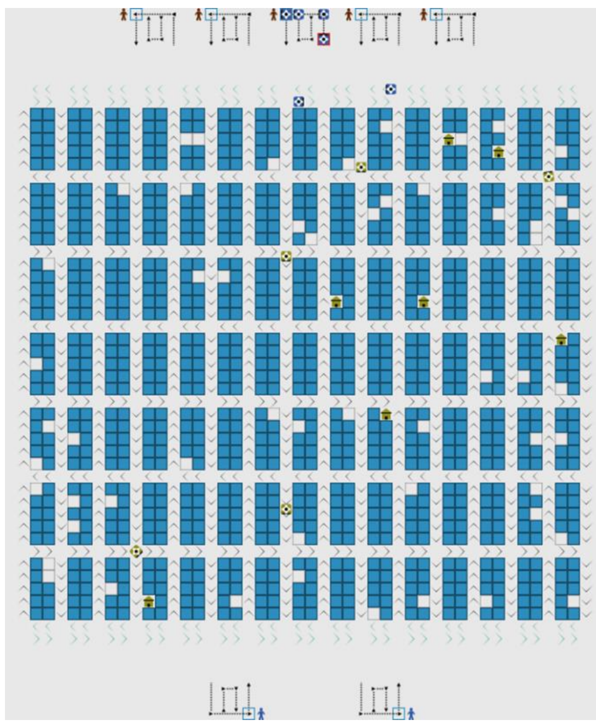


Figure 3.5 Simulation Layout

3.3.3 Simulation Parameter

In terms of simulation, several parameters are used to simulate the entire process of RMFS. The parameters and assumptions used for simulation in this research can be seen in Table 3.1.

Table 3.1 Simulation Parameter Setting

Parameter	Value
Simulation	
Run Length	8 hours
Replication	30 replications for each scenario
Layout	

Parameter	Value
Inventory Area	1050 locations
Inventory Capacity	999 pods
Empty Storage Area	51 locations
Pod Batch	2 x 5 blocks
Aisles	16 vertical aisles 8 horizontal aisles
Stations	5 picking stations 2 replenishment stations
	7 charging stations
Orders	
Order Line	Multi-line orders
Initial Orders	100 orders
Pods	
Capacity	100 units
SKUs	
Number of SKUs	8939 SKUs
SKU Distribution	Class A = 0-893 (10%) SKUs Class B = 894-3575 (30%) SKUs Class C = 3576-8938 (60%) SKUs
SKUs/pod	20 SKUs (ABC Mixed Class Storage Policy)
AGVs	
Number of AGVs	50 AGVs
AGV Speed	1.5 m/s (load), 2 m/s (no load)
AGV Acceleration/Deceleration	1 m/s
Time to lift and store pod	4 second
AGV to pod policy	Shortest Pod
Picking	
Queuing per station	12 AGVs
Picking Time	10
Picking Policy	Random Similarity-Based Order Batching and Order Assignment
Replenishment	
Queuing per station	6 AGVs
Replenishment time	20
Warehouse Inventory Level	60%
Pod Inventory Level	90%

The simulation is conducted for eight hours with thirty replications. The replication technique is used to lessen the influence of randomness. The warehouse layout is shown in the preceding section, and the exact parameter settings are presented in the table above. This study utilizes multi-line orders as input for order data and implements the ABC class rule to SKU classification. The SKU-to-Pod distribution proportionately assigns each class to each pod. One pod may only include a maximum of 20 different SKUs, and the pod's unit distribution is as follows: 60% Class A units, 25% Class B units, and 15% Class C units. When it comes to replenishment, a trigger will be set off either when the inventory level of the warehouse falls below 60% or when the inventory level of an individual pod falls below 90%.

3.4 Pick Order Assignment Problem Formulation

This research will examine two problems in warehouse operations: order batching and pick-order assignment. This study will use a simple mathematical model to represent the system and use system simulation to replicate the real warehouse environment. The simulation will enable us to incorporate complexities and uncertainties that may be difficult to represent in a purely mathematical model. It will also allow us to capture the system's dynamic behavior accurately. Furthermore, the use of simulation allows for the testing of different scenarios: With simulation, it is easy to test the behavior of a system under various conditions or scenarios. In the following section, we will provide a more detailed explanation of these problems for each scenario.

3.4.1 Order Batching Model

The order batching model represents how the system will assign some orders to the same pick order list. Below is the explanation of the order batching model for the two scenarios.

1. Baseline Model

   As explained in the literature review, the previous studies focused on deciding the optimal number batch size or batching time in a static environment. Furthermore, previous studies did not consider the rule to assign which order needs to be grouped into the same batch. Then, for the baseline model, the order batching will apply the random rule to allocate the order into the same batch. This random rule is used as a baseline for comparison with the proposed model that uses the similarity rule.

2. Proposed Model

In the proposed model, the order grouping model classifies orders into batches based on the similarity SKU types, as measured using the Jaccard Index. After calculating the similarity of the orders, the model follows a clustering approach to group orders into batches, with the number of picking stations serving as the number of batches required. The formulation for measuring order similarity and grouping orders is shown in Equations 3.1 to 3.4.

$$Sim_{AB} = \frac{C}{|I|} x \frac{C}{|J|} \tag{3.1}$$

Notations:

$I\ (i, \dots, I)$	: Set of SKUs contain in order A
$J\ (j, \dots, J)$	: Set of SKUs contain in order B
x_{ij}	: $\begin{cases} 1, if\ i \in I\ \cap j \in J \\ 0, otherwise \end{cases}$
$C = \sum_{i=1}^{I} \sum_{j=1}^{J} x_{ij}$	: Set of SKUs contains both order A and order B

Objective

$$Max \sum_{i=1}^{P} \sum_{j=1}^{K} S_{ij} X_{ij} \tag{3.2}$$

s.t

$$\sum_{j=1}^{K} X_{ij} = 1 \qquad , \forall i \in \{i, \dots, P\} \tag{3.3}$$

$$\sum_{i=1}^{P} Y_j = K \tag{3.4}$$

Notations:

P	: Total number of orders
K	: Number of picking stations
x_{ij}	: $\begin{cases} 1, if\ order\ i\ is\ assigned\ to\ group\ j \\ 0, otherwise \end{cases}$
y_j	: $\begin{cases} 1, if\ order\ j\ is\ selected\ as\ a\ group\ center \\ 0, otherwise \end{cases}$

For the order batching model, equation (3.1) represents order similarity calculation between two orders. Equation (3.2) is the objective that maximizes the order similarity of each order inside a batch. Constraint (3.3) ensures that each order can only be assigned to one picking station. Furthermore, constraint (3.4) represents that the batch number cannot exceed the number of picking stations.

Based on the model above, the number of batches will be the same as the number of picking stations in the warehouse. Each order that arrived in the system during a specific time will be grouped based on their SKU's similarity to a picking station. In a static system, we can predict all the orders in advance, making it easier to distribute batches according to a mathematical model. However, using a mathematical model to represent a dynamic system where orders keep coming in over time is difficult. This research proposes adjusting the settings of the batching process to define the system's dynamic behavior. Precisely, the number of batches will be adjusted based on the number of idle picking stations at a given time, and the number of orders assigned to each batch will be adjusted based on the available capacity of the slot picking stations. This approach is more representative of the actual situation in the RMFS warehouse, where orders are immediately processed and exit the system after being picked. Additionally, this approach increases the system's utility by avoiding idle slots and allowing for continuous processing of orders rather than waiting for all orders to be completed before moving on to the next assignment. The proposed model allows the system to adjust the number of batches and the number of orders in each batch as needed.

3.4.2 Order Assignment Problem

The order assignment model represents how the system will assign each order inside a batch to a specific picking station. Then, the system will assign which pod can fulfill the order being processed in a particular working station. The main objective of this model is to minimize the number of pods needed to be delivered to the picking station for fulfilling the whole order in a specific time. The detailed order assignment scenarios are described as follows.

1. Baseline

   Regarding order assignment, the baseline model uses an arbitrary rule to assign the order to the designated picking station. Then, the system will assign the most pile-on pods to fulfill the order and deliver it to the designated picking station. The detailed order assignment process of the baseline scenario shows in Figure 3.6.

2. Proposed Model

The model in Equations 3.5 represents the order assignment process in the proposed model to 3.15

Objective

$$Min \sum_{i=1}^{P} \sum_{j=1}^{K} Y_{jkt} \tag{3.5}$$

s.t

$$\sum_{j=1}^{m} \sum_{k=1}^{l} Z_{ijsk} \geq O_{is}, \quad \forall i \in \{1, \ldots, n\} \ \forall s \in \{1, \ldots, \}, \tag{3.6}$$

$$\sum_{k=1}^{l} W_{ik} \leq 1, \quad \forall i \in \{1, \ldots, n\} \tag{3.7}$$

$$\sum_{t=1}^{r} \sum_{i=1}^{n} D_{ikt} \leq C, \quad \forall k \in \{1, \ldots, t\} \tag{3.8}$$

$$\sum_{k=1}^{l} Y_{jkt} \leq 1, \quad \forall k \in \{1, \ldots, t\} \tag{3.9}$$

$$Y_{jkt}.R_{js} \geq Z_{ijsk} \tag{3.10}$$

$$H_{it} = D_i \times AVG \tag{3.11}$$

$$D_{ikt} = \begin{Bmatrix} 0, ST_i > T \\ 1, otherwise \end{Bmatrix} \tag{3.12}$$

$$ST_i = \begin{Bmatrix} 0, i \leq C \\ FT_{i-1}, otherwise \end{Bmatrix} \tag{3.13}$$

$$FT_i = \begin{Bmatrix} H_{it}, i \leq C \\ FT_{i-1} + H_{it}, otherwise \end{Bmatrix} \tag{3.14}$$

$$\sum_{i=1}^{n} H_{it}.W_{ik}.D_{ikt} \leq T \quad \forall k \in \{1, \ldots, t\} \tag{3.15}$$

Notations

$i = 1, \ldots, n$	: Set of orders
$j = 1, \ldots, m$	: Set of pods
$s = 1, \ldots,$ p	: Set of SKUs
$k = 1, \ldots, l$	: Set of picking station
$b = 1, \ldots, u$	: Set of batches
$t = 1, \ldots, T$	
C	: Capacity of the picking station

Demand	: Order Quantity
AVG	: Average picking time per SKU
O_{is}	: $\begin{cases} 1, if\ order\ i\ include\ SKU\ s \\ 0, otherwise \end{cases}$
R_{js}	: $\begin{cases} 1, if\ pod\ j\ include\ SKU\ s \\ 0, otherwise \end{cases}$
W_{ik}	: $\begin{cases} 1, if\ order\ i\ is\ assigned\ to\ picking\ station\ k \\ 0, otherwise \end{cases}$
Y_{jkt}	: $\begin{cases} 1, if\ pod\ j\ need\ to\ be\ delivered\ to\ station\ k \\ 0, otherwise \end{cases}$
Z_{ijsk}	: $\begin{cases} 1, if\ SKUs\ required\ for\ order\ i\ delivered\ by \\ pods\ j\ in\ station\ k \\ 0, otherwise \end{cases}$
D_{ikt}	: $\begin{cases} 1, if\ order\ i\ is\ processed\ in\ station\ k \\ in\ spesific\ time\ t \\ 0, otherwise \end{cases}$

$H_{it} = Demand(i).AVG$

Equation (3.5) is the objective of the order assignment model, which aims to minimize the number of pods needed to fulfill the order in a specific time. Constraint (3.6) ensures that all the SKUs need to be fulfilled using all the warehouse resources. Constraint (3.7) states that one order only can be assigned to one picking station. Constraint (3.8) means that the number of orders processed consistently in the picking station cannot exceed the capacity of the picking station. Constraint (3.9) ensures that one pod can only be delivered to one picking station. Equation (3.10) makes sure that if a pod is assigned to the picking station, it means that the pod can fulfill the SKUs required in station k. Equation (3.11) represents the processing time for each order. Equation (3.12) decides whether an order will be processed or not. Equations (3.13) and (3.14) represent each order's starting and finish times. Last, equation (3.15) ensures that the total service time of all orders being processed is not more than the maximum time for order picking fulfillment.

The detailed order assignment process of the baseline scenario shows in Figure 3.7. The system will start when the system receives an order and then check if there is sufficient inventory for each order. If the order can be fulfilled with the current system inventory, then the order can be processed by the system for the assignment part. Otherwise, if there is no sufficient quantity or SKU in the pod for an order, the order will be kept in the order pool. Then, the system will identify which picking station is available for the new assignment, how many picking stations are, and how much capacity is available for placing a new order assignment. Then, the system will identify the bin and station availability in the assignment process.

Figure 3.6 System Flowchart for Baseline Model

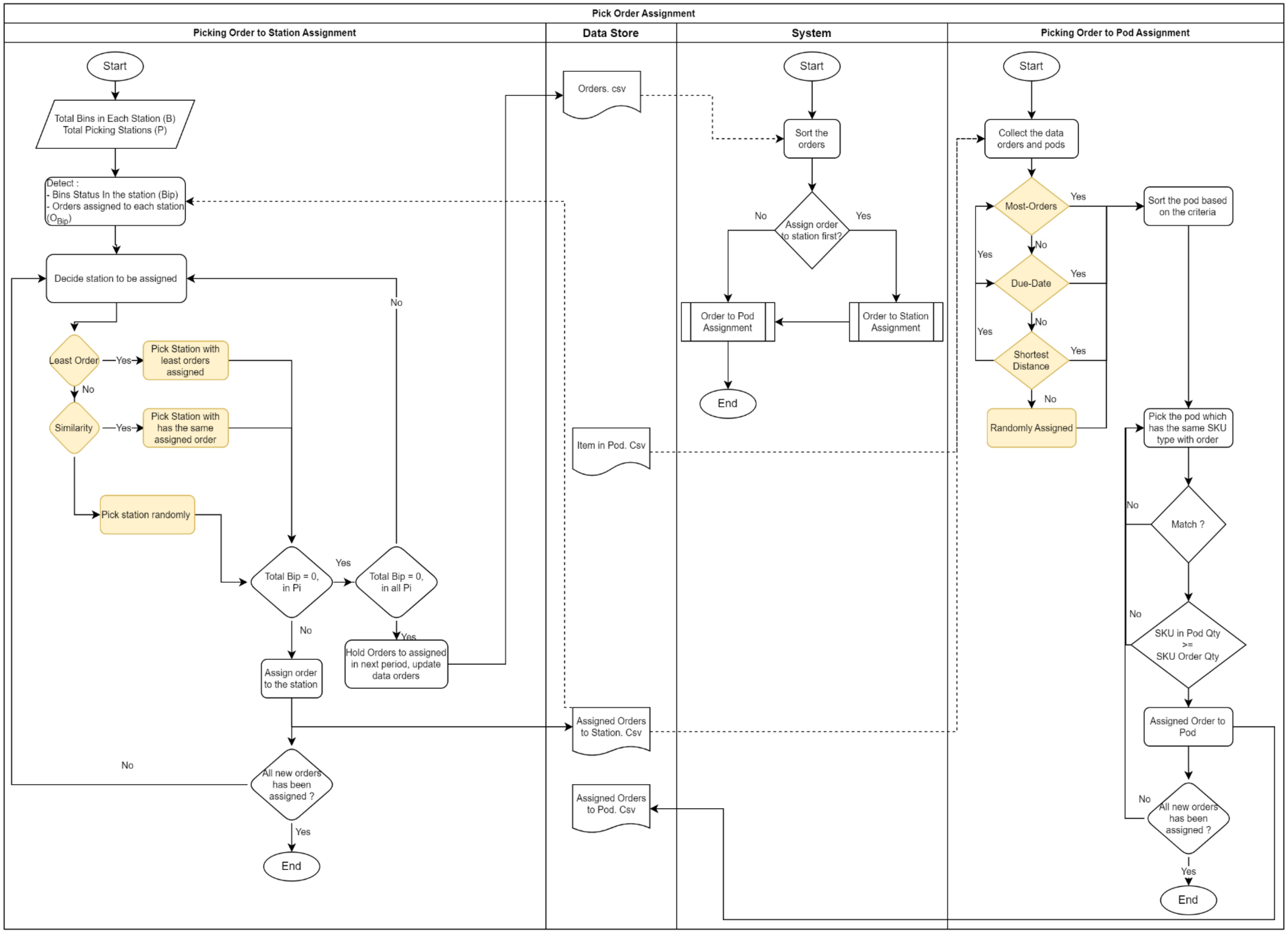
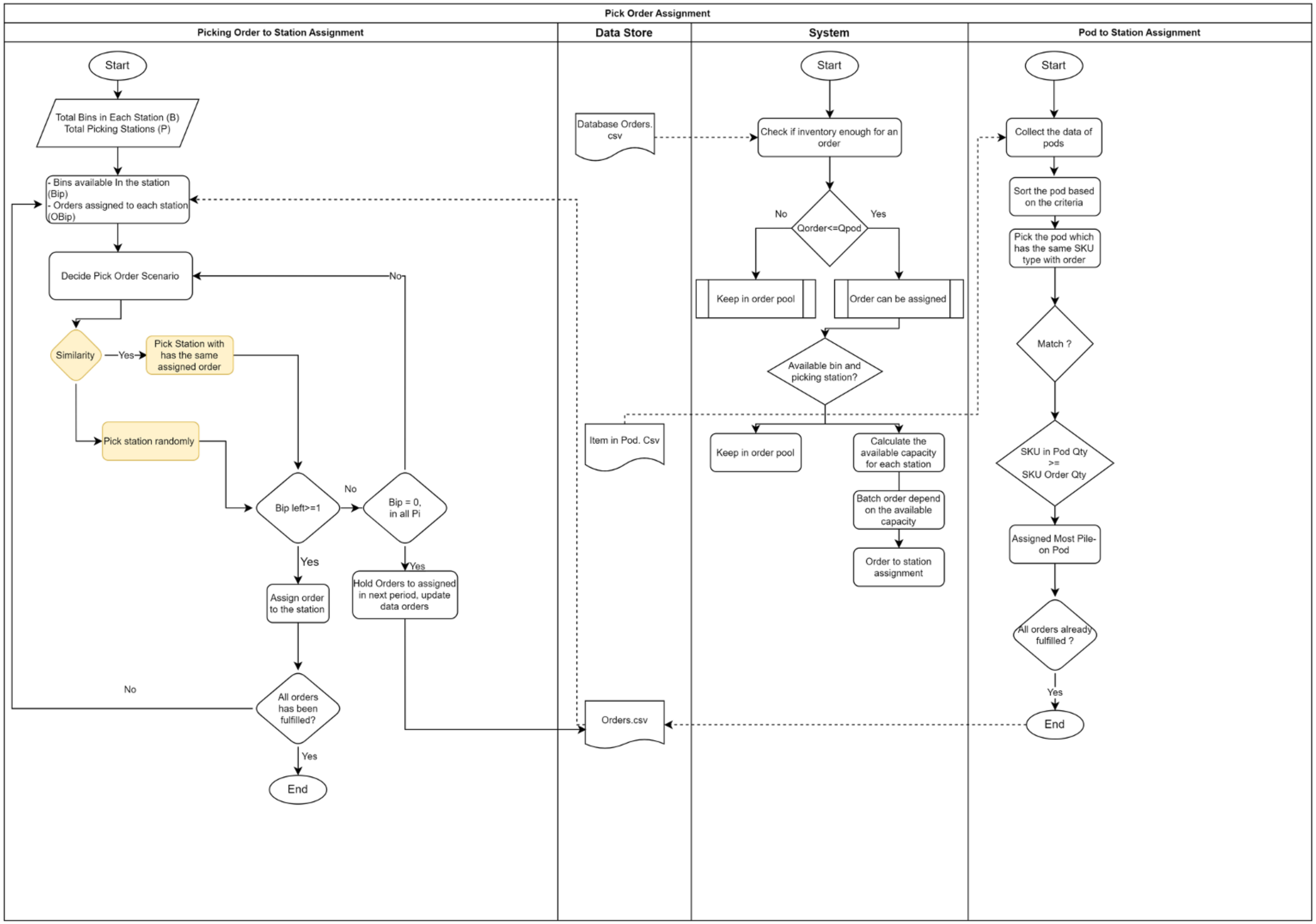


Figure 3.7 System Flowchart for Proposed Model



Then, the assignment will be run depending on the user statement of which scenario will be used, whether random or similarity-based. The order that needs to assign every time is not only one order. More than one order might be assigned for each picking station. So, this model also specifies which order should be assigned to which picking station and the sequence.

3.5 Performance Analysis

System performance evaluation will involve comparing simulation results of different scenarios using a statistical test. Furthermore, a sensitivity analysis experiment is conducted with several parameter changes to examine the best scenario's robustness.

3.5.1 Evaluation Criteria

In order to evaluate the performance of the warehouse simulation, this research uses several indicators to compare the impacts of different scenarios resulting from the simulation. Notations used in the performance measurement criteria can be seen as follows.

Notations

OR_t : Order rate in period *t*

TR_t : Throughput rate in period *t*

TE_t : Throughput efficiency in period *t*

P : Total number of pods

np_t : Number of picked pods in period *t*

P_{np} : Percentage of the number of pods used

po : Pile-on

The performance criteria used in this research are described below.

1. Pile-on Value

   Pile-on value means the average number of order items picked in one pod. It is calculated by throughput rate (TRt) and the number of picked pods (npt) in period *t*. Equation (3.16) is shown below.

$$po = \frac{TR_t}{np_t} \tag{3.16}$$

2. Percentage of the number of pods transported in a specific period t

The percentage of the number of pods transported is the ratio of the number of pods used denotes how many pods be picked (npt) within the total number of pods (P). The equation is shown as follows.

$$P_{np} = \frac{np_t}{P} \text{ x } 100 \qquad (3.17)$$

This indicator can reflect the objective of the system optimization. For example, the lower percentage indicates reduced robot movement and shows better performance.

3. *Throughput Efficiency*

   Throughput efficiency can be calculated by dividing order rate (ORt) and throughput rate (TRt) in period *t* as follows.

$$TE_t = \frac{TR_t}{OR_t} \qquad (3.3)$$

Order rate is the total number of arriving orders in the system at period *t,* while throughput shows how many orders can be fulfilled in period *t*. Based on this statement, we know that throughput efficiency represents how effectively the orders can be finished within period *t*. the higher throughput efficiency means the better performance of the system.

In this research, the simulation is run 30 replications for each scenario. These replications are used because there are randomness factors inside the simulation system. Replication aims to know the simulation's reliability and robustness by evaluating the simulation results' standard deviation. It is also used to observe the randomness and deviation of the model. The lower deviation indicates that the model is more robust and can produce results with few differences during several replications. In order to evaluate the difference between two scenarios in a statistical way needs, a minimum of 30 times the results for each scenario.

This research aims to increase the total throughput by maximizing the pile-on and minimizing the total number of pods needed for fulfilling the order. The pile-on number represents a pod’s levels of utility since a high pile-on value indicates that a pod can fill more SKUs in a single trip. Because the more excellent the pile-on value means, the more likely the order list may be fulfilled faster. Generally, the pile-on value is also related to the number of pods required to satisfy orders. This is because the larger the pile-on number, the fewer pods will likely be needed to finish the order list. As a result, there were fewer pods transported to the picking station. Then the AGVs did not have to move as much throughout the warehouse. Besides that, fewer pods delivered must optimize pod use to ensure warehouse throughput efficiency.

3.5.2 Sensitivity Analysis

Sensitivity analysis is used to measure the robustness and reliability of the model with specific changes in the parameter. In the simulation, sensitivity analysis is frequently used to test the robustness and reliability of the results. Simulation is a powerful method for investigating and predicting complex systems' behavior, but the results' accuracy can be affected by the model's assumptions and input parameters.

Sensitivity analysis can help identify which input parameters affect results most and can be used to evaluate the simulation's robustness to changes in these parameters. In this research, the sensitivity analysis is conducted by modifying several parameters. Sensitivity analysis also can show how much the parameter can be changed without affecting the objective value. This research performs the sensitivity analysis for the proposed model as follows.

1. The number of the picking station
2. Number of picking station capacity
3. Batching Time
4. Number of AGVs

CHAPTER 4

RESULTS AND DISCUSSION

This chapter will explain and show the research process starting from data generation, data processing, and results analysis.

4.1 Data

This section will explain in more detail the data used in this research. Several datasets are needed to run the simulation: database orders, orders, items in pods, and order finished dataset.

4.1.1 Data Variables

The dataset used in this research is generated warehouse dataset. The data are generated following specific parameters. Database orders consist of several variables, as listed in Table 4.1.

Table 4.1 Order Database Characteristics

Category	Column Name	Description	Data Value
Database Orders	order_id	The sequence of the order after the order created in the simulation	Float
	order_dum	The sequence of the order, which represent the order_id of **database_order**	Float
	order_type	Differentiate the order type, picking & replenishment	[1] picking [2] replenishment
	item	Product/ SKU ID	[0, 1, 2, ...]
	quantity	Quantity of SKUs that are being ordered	Integer
	facing	The facing of the pod where the order being assigned	[-1], no order assigned [0, 1]. order assigned
	due_date	Order due date	"[-1], no order assigned Float, order assigned."
	station	The station that chooses based on the order	"[-1], no order assigned [0, 1, 2, ...]. order assigned"
	pod_id	The pod that gets assigned by the order	"[-1], no order assigned [0, 1, 2, ...]. order assigned"
	status	Order Status	
	finish_time	The time when the order is already being fulfilled	"[-1], no order assigned [0, 1, 2, ...]. order assigned"
	date	The date of the order	Integer
	time_gen	The time when the order was generated on that date	Integer

The database order records all of the demands from customers with all of the details characteristics above.

4.1.2 Orders Data

In this research, the simulation will be run for 8 hours. Order data will always be automatically updated and receive data from the order database as time goes on in the simulation. There are 1803 orders included in the system. The order has variable characteristics, as listed in Table 4.2

Table 4.2 Orders Data Characteristics

Category	Column Name	Description	Data Value
Orders / Orders Finish	order_id	The sequence of the order after the order created in the simulation	Float
	order_dum	The sequence of the order, which represent the order_id of **database_order**	Float
	order_type	Differentiate the order type, picking & replenishment	[1] picking [2] replenishment
	item	Product/ SKU ID	[0,1, 2, ...]
	qty	Quantity of SKUs that are being ordered	Integer
	facing	The facing of the pod where the order being assigned	[-1], no order assigned [0, 1], order assigned
	due_date	Order due date	"[-1], no order assigned Float, order assigned."
	station	The station that chooses based on the order	"[-1], no order assigned [0, 1, 2, ...]. order assigned"
	pod_id	The pod that gets assigned by the order	"[-1], no order assigned [0, 1, 2, ...]. order assigned"
	status	Order Status	-3: the order has not been assigned -2: the order has not been assigned to the pod but has already been assigned to the station -1: order assigned but has not been picked by AGV 0: order assigned and picked by AGV 1: order finished
	finish_time	The time when the order is already being fulfilled	"[-1], no order assigned [0, 1, 2, ...]. order assigned"

4.1.3 Item in Pod Data

Item in pod data is the dataset that contains the list of all items, also called SKUs, in each pod for all pods in the warehouse. Item in pod data is generated using a specific rule. The parameter setting for generating this dataset can be seen in Table 4.3, and the generated item in pod data has several variables, as listed in Table 4.4.

Table 4.3 Item in Pod Data Generation

SKUs	
Number of SKUs	8939 SKUs
SKU Distribution	Class A = 0-893 (10%) SKUs
	Class B = 894-3575 (30%) SKUs
	Class C = 3576-8938 (60%) SKUs
Slots/pod	20 SKUs

Table 4.4 Item in Pod Data Characteristics

Category	Column Name	Description	Data Value
Item in Pod	pod_id	Identification of the pod that is being used in the warehouse	[0] to [Total Pod - 1]
	pod_type	Differentiate the type of pod being used. If there is only 1 type, the value is 0.	[0, 1, 2, ……]
	slot_id	Represent the slot type (based on volume) **used in a pod**. Where the facing of the pod has the same slot design	1. only 1 pod, the value is [0, 1, 2, …….] 2. more than 1 pod. Pod 1 [0, 1, 2, 3, 4, 5] Pod 2 [6, 7, 8, 9, 10]
	item	Product/ SKU IDTotal	[0, 1, 2, ...]
	qty	Quantity left of SKU stored in the slot	Integer
	max_qty	The maximum quantity of SKUs stored in the slot	Integer
	due_date	Expired/ due date of the SKU	Float
	facing	Facing the side of the pod	[0, 1]
	pick_ind	Index if the slot being assigned by an order	0, no order being assigned 1, the order is assigned

4.2 Experimental Results

This subchapter is divided into two parts: the experiment results for each scenario and the comparison of the two scenarios.

4.2.1 Simulation Scenarios Results

The simulation process in this research is carried out using the NetLogo simulation platform with specific parameter settings, as written in section 3.3. The simulation will be run for 8 hours with 30 replications. The simulation results here will be differentiated into the results of the baseline scenario and the proposed scenario.

1. Baseline Scenarios

   This scenario will follow the random rule for allocating orders in the same batch. This scenario does not consider how to group the orders into the same batch. Furthermore, this scenario randomly assigns the order to the picking station following the first come, first serve rule. The system will assign the pods to a picking station based on the most pile-on rule. The results of baseline scenarios are summarized in

   Table 4.5.

Table 4.5 Baseline Scenario Results

Replication	**Performance Measure**			
	Number of pods (np_t)	**% Number of pods (P_{np})**	**% Throughput Efficiency (TE_t)**	**Pile-on (po)**
1	543	54.35	92.38	2.39
2	551	55.16	94.76	2.36
3	541	54.15	93.33	2.41
4	543	54.35	93.81	2.39
5	553	55.36	94.05	2.35
6	539	53.95	94.76	2.39
7	541	54.15	93.57	2.38
8	543	54.35	93.81	2.39
9	537	53.75	93.10	2.40
10	545	54.55	94.05	2.39
11	548	54.85	92.14	2.36
12	533	53.35	92.38	2.41
13	547	54.75	93.10	2.38
14	545	54.55	91.90	2.39
15	549	54.95	90.48	2.36
16	531	53.15	92.38	2.43
17	544	54.45	94.29	2.39
18	558	55.86	91.43	2.33
19	545	54.55	89.29	2.37
20	533	53.35	93.57	2.44
21	544	54.45	94.29	2.39
22	537	53.75	92.14	2.39
23	540	54.05	92.62	2.41

Replication	Performance Measure			
	Number of pods (np_t)	% Number of pods (P_{np})	% Throughput Efficiency (TE_t)	Pile-on (po)
24	543	54.35	90.71	2.40
25	538	53.85	92.86	2.39
26	547	54.75	93.10	2.37
27	538	53.85	90.95	2.41
28	548	54.85	93.57	2.38
29	548	54.85	94.05	2.36
30	538	53.85	89.76	2.40

2. Proposed Scenarios

The proposed scenario uses similarity-based order batching rule that allows the system to assign orders with higher similarity into the same batch and assign it to the same picking station. In order assignment, this scenario will assign the batch of orders to the picking station with the highest similarity or active orders and the orders inside the batch Table 4.6.

Table 4.6 Proposed Model Scenario Results

Replication	Performance Measure			
	Number of pods (np_t)	% Number of pods (P_{np})	Throughput Efficiency (TE_t)	Pile-on (po)
1	324	32.43	96.41	4.05
2	323	32.33	94.74	4.01
3	330	33.03	94.50	3.95
4	323	32.33	94.74	4.03
5	331	33.13	95.93	3.93
6	321	32.13	95.22	4.05
7	330	33.03	96.89	3.96
8	303	30.33	98.09	4.41
9	333	33.33	94.50	3.90
10	324	32.43	95.69	4.05
11	328	32.83	95.69	3.97
12	330	33.03	94.50	3.94
13	336	33.63	98.09	3.91
14	326	32.63	94.74	4.01
15	331	33.13	93.30	3.91
16	329	32.93	94.26	3.95
17	326	32.63	94.26	3.98
18	329	32.93	95.22	3.93
19	344	34.43	95.45	3.78
20	341	34.13	95.45	3.57
21	301	30.13	97.61	4.30
22	319	31.93	95.93	4.29

Replication	Performance Measure			
	Number of pods (np_t)	% Number of pods (P_{np})	Throughput Efficiency (TE_t)	Pile-on (po)
23	349	34.93	96.65	3.74
24	215	21.52	99.28	6.20
25	326	32.63	94.74	4.14
26	343	34.33	95.22	3.72
27	332	33.23	97.61	4.03
28	316	31.63	96.65	4.01
29	336	33.63	94.50	3.68
30	306	30.63	95.45	3.91

4.2.2 Scenarios Comparison

Based on the simulation results, each scenario is simulated for 30 replications. The comparison of the scenarios is listed in Table 4.7.

Table 4.7 Scenario Results Comparison

Replication	Performance Measure							
	Number of pods (np_t)		% Number of pods (P_{np})		Throughput Efficiency (TE_t)		Pile-on (po)	
	Baseline	*Proposed*	*Baseline*	*Proposed*	*Baseline*	*Proposed*	*Baseline*	*Proposed*
1	543	324	54.35	32.43	92.38	96.41	2.39	4.05
2	551	323	55.16	32.33	94.76	94.74	2.36	4.01
3	541	330	54.15	33.03	93.33	94.50	2.41	3.95
4	543	323	54.35	32.33	93.81	94.74	2.39	4.03
5	553	331	55.36	33.13	94.05	95.93	2.35	3.93
6	539	321	53.95	32.13	94.76	95.22	2.39	4.05
7	541	330	54.15	33.03	93.57	96.89	2.38	3.96
8	543	303	54.35	30.33	93.81	98.09	2.39	4.41
9	537	333	53.75	33.33	93.10	94.50	2.40	3.90
10	545	324	54.55	32.43	94.05	95.69	2.39	4.05
11	548	328	54.85	32.83	92.14	95.69	2.36	3.97
12	533	330	53.35	33.03	92.38	94.50	2.41	3.94
13	547	336	54.75	33.63	93.10	98.09	2.38	3.91
14	545	326	54.55	32.63	91.90	94.74	2.39	4.01
15	549	331	54.95	33.13	90.48	93.30	2.36	3.91
16	531	329	53.15	32.93	92.38	94.26	2.43	3.95
17	544	326	54.45	32.63	94.29	94.26	2.39	3.98
18	558	329	55.86	32.93	91.43	95.22	2.33	3.93
19	545	344	54.55	34.43	89.29	95.45	2.37	3.78
20	533	341	53.35	34.13	93.57	95.45	2.44	3.57
21	544	301	54.45	30.13	94.29	97.61	2.39	4.30
22	537	319	53.75	31.93	92.14	95.93	2.39	4.29
23	540	349	54.05	34.93	92.62	96.65	2.41	3.74
24	543	215	54.35	21.52	90.71	99.28	2.40	6.20

Replication	Performance Measure							
	Number of pods (np_t)		% Number of pods (P_{np})		Throughput Efficiency (TE_t)		Pile-on (po)	
	Baseline	***Proposed***	***Baseline***	***Proposed***	***Baseline***	***Proposed***	***Baseline***	***Proposed***
25	538	326	53.85	32.63	92.86	94.74	2.39	4.14
26	547	343	54.75	34.33	93.10	95.22	2.37	3.72
27	538	332	53.85	33.23	90.95	97.61	2.41	4.03
28	548	316	54.85	31.63	93.57	96.65	2.38	4.01
29	548	336	54.85	33.63	94.05	94.50	2.36	3.68
30	538	306	53.85	30.63	89.76	95.45	2.40	3.91
Average	543	324	54.35	32.38	92.75	95.71	2.39	4.04
Gap	**40.33%**		**40.42%**		**-3.19%**		**-69.03%**	

Based on the results above, we can say that the proposed model can significantly lower the number of pods needed to fulfill the order. As written in Table 4.7, the proposed model outperforms and is competitive enough compared with the baseline scenario. Using similarity-based order batching and the order assignment model can decrease the average number of pods needed by 40%. It is in line with the pods' utilization percentage, which means the number of pods needed to be divided by the total number of pods in the warehouse.

Furthermore, putting the order with the highest similarity into the same batch has proven that it can increase the pile-on value of the system since it allows the simulation to achieve almost 70% better pile-on compare with the baseline scenario. The proposed model also sequences the order assignment to the station based on the similarity, allowing a higher chance of similar orders being processed concurrently. This strategy helped the system achieve higher throughput of around 3% by assigning similar orders concurrently within the batch and considering the active order in the picking station before assigning the batch. This order assignment strategy increases the possibility and ensures that every order and/or batch of orders always goes to the picking station with the most similar order. Then, it will help to maximize the pile-on, minimize the pod movement, and decrease the service time for each order fulfillment.

4.3 Statistical Analysis

Statistical analysis is used to compare the difference between the two scenarios. The One-way *t-test* test is used in this research to analyze the mean average difference between the two scenarios. A *t-test* test is used to compare and examine the difference between two groups by comparing the average mean between two scenarios. Before doing the *t-test*, some assumptions need to be fulfilled. In order to use the *t-test* test, the data should satisfy the normality assumption and have equal variance. This research uses Shapiro-Wilkin Test to examine the normality of the data. Furthermore, the Levene test is conducted to test the equality of variance between groups.

The normality test is conducted following the hypothesis as written in Table 4.8. Then, conversely, the equality variance test follows the hypothesis as written in Table 4.9.

Table 4.8 Hypothesis for Normality Test

H_o	All means are equal
H_1	Not all means are equal
Significance level	$\alpha = 0.05$

Table 4.9 Hypothesis for Equality Variance Test

H_o	All variances are equal
H_1	At least one variance is different
Significance level	$\alpha = 0.05$

Using Minitab software, the normality test and equality variance test results for each evaluation criterion are listed in Table 4.10.

Table 4.10 The P-value Result

Data	Shapiro-Wilk	Kolmogorov-Smirnov	Levene-Test
Percentage Number of Pods (baseline)	> 0.10	> 0.15	0.085
Percentage Number of Pods (proposed)	< 0.01	< 0.01	
Throughput Efficiency (baseline)	> 0.10	> 0.15	0.021
Throughput Efficiency (proposed)	0.09	0.128	
Pile-on (baseline)	> 0.10	< 0.01	0.077
Pile-on (proposed)	< 0.01	< 0.01	

Table 4.10 specifies the p-value for each group of data. Based on the p-value, not all data satisfy the normality test and have equal variance.

- The percentage number of pods in the baseline scenario follows normal distribution since the p-value for both normality tests is over the significance level of 0.05. On the other hand, the percentage number of pods in the proposed scenario does not satisfy the normality assumption since the p-value for both tests is lower than the significance level. Furthermore, the Levene test showed that both groups have equal variance. Based on this situation, because one of the assumptions is not satisfied, this research will use Wilcoxon Sum-Ranked test as a nonparametric test replacing the *t-test*.

- The normality test for the throughput efficiency in both scenarios proves that both groups satisfy the normality assumption since their p-value is more than the significance level. Then, regarding the Levene test, there is equal variance between the baseline model's and the proposed model's throughput efficiency. Moreover, in this case, the *t-test* test will be conducted to examine the average mean difference between the two groups.
- The p-value for pile-on data of the baseline scenario showed the difference between Kolmogorov-Smirnov and Shapiro-Wilkin tests. Based on the Shapiro-Wilkin, the data follow the normal distribution, but the Kolmogorov-Smirnov test proves that the data do not satisfy the normality assumption. On the other hand, the pile of data of the proposed model presents the same result that the data follow the normal distribution. Regarding the equal variance test, the pile on the data did not have equal variance since the p-value for the Levene test was more than 0.05. Based on this situation, because none of the parametric assumptions is satisfied, this research will use the Wilcoxon Ranked-Sum test as a nonparametric test replacing the *t-test*.

Based on the assumption test above, the Wilcoxon Rank-Sum test will be conducted to compare the average mean difference between the percentage number of pods used and Pile-on between the baseline and proposed scenario. Then, the *t-test* is used to examine the mean difference in throughput efficiency criteria between both scenarios.

1. Percentage Number of Pods Needed

   Wilcoxon Ranked-Sum for comparing the percentage number of pods used of baseline and proposed scenario.

Test

Null hypothesis	$H_0: \eta_1 - \eta_2 = 0$
Alternative hypothesis	$H_1: \eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	1365.00	0.000
Adjusted for ties	1365.00	0.000

Figure 4.1 Wilcoxon Rank-Sum Test Result for Percentage Number of Pods Needed

As shown in Figure 4.1, we can see that the p-value is less than the significance level of 0.05. So, we can 95% say there is a significant difference between the percentage number of pods used between the baseline and the proposed scenario.

2. Percentage of Throughput Efficiency

   the t-test is utilized to test the average mean of percentage throughput efficiency between the baseline and proposed model.

Test

Null hypothesis	H_0: $\mu_1 - \mu_2 = 0$
Alternative hypothesis	H_1: $\mu_1 - \mu_2 \neq 0$

T-Value	DF	P-Value
-8.11	58	0.000

Figure 4.2 t-test Result for Percentage of Throughput Efficiency

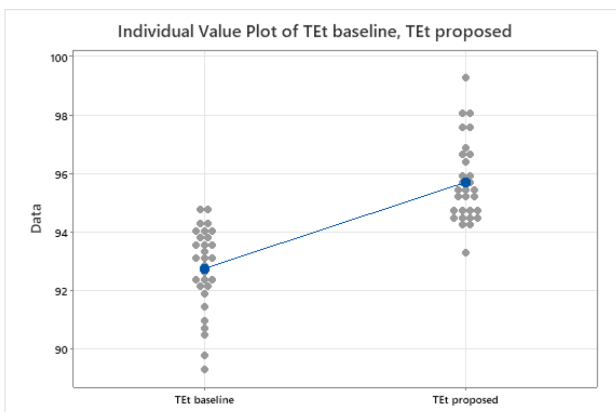


Figure 4.3 Order Picking System Classification

Based on the result shown in Figures 4.2 and 4.3, we can see that the p-value is less than the significance level of 0.05. So, we can 95% say there is any significant difference between the percentage of throughput efficiency used between the baseline and the proposed scenario.

3. Pile on Value

   Wilcoxon-Rank Sum test is utilized to test the average pile-on value between the baseline and proposed model.

Test

Null hypothesis H_0: $\eta_1 - \eta_2 = 0$
Alternative hypothesis H_1: $\eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	465.00	0.000
Adjusted for ties	465.00	0.000

Figure 4.4 Wilcoxon Rank-Sum Test Result for Pile-on Value

Figure 4.4 shows that the p-value is less than the significance level of 0.05. So, we can 95% say there is a significant difference between the percentage number of pods used between the baseline and the proposed scenario.

All results prove a significant difference between the baseline and the proposed model for all evaluation criteria. It shows that the proposed model is competitive enough to improve the performance of the order-picking warehouse. Moreover, these results are only based on specific parameter settings. In order to examine the reliability, robustness, and consistency of the model, this research uses sensitivity analysis, which will be explained more in the next chapter.

4.4 Sensitivity Analysis

In this study, different parameter values might affect the system's results. Moreover, the sensitivity analysis is used to identify how much variations in the input values for a given variable impact the results for the proposed model. The sensitivity analysis is conducted for four different parameters: the number of picking stations, picking station capacity, batching time, and number of AGVs. The sensitivity is evaluated by changing one parameter and keeping the other parameter as the default proposed model's parameter. The detailed results are listed in the following Table 4.11. Then, the performance comparison for each parameter depicts in Figure 4.5 to Figure 4.8.

Table 4.11 Sensitivity Analysis Results Comparison

Parameter	Value	Performance Measure			
		Number of pods (np_t)	% Number of pods (P_{np})	Throughput Efficiency (TE_t)	Pile-on (po)
Picking Station	2	250	25.03	57.89	3.72
	3	295	29.53	69.38	3.64
	4	328	32.83	93.30	2.39
	5	543	32.43	95.93	4.01
	6	326	32.63	96.41	4.00
Station Capacity	8	323	32.33	94.50	4.04

Parameter	Value	Performance Measure			
		Number of pods (np_t)	% Number of pods (P_{np})	Throughput Efficiency (TE_t)	Pile-on (po)
	12	332	33.23	93.30	3.90
	16	543	32.43	95.93	4.01
	20	326	32.63	94.26	3.98
	24	320	32.03	96.41	4.08
Batching Time	5	324	32.43	95.93	4.01
	10	315	31.53	80.38	3.76
	15	240	24.02	48.09	3.75
	20	259	25.93	44.74	3.47
	25	225	22.52	33.73	3.34
Number of AGVs	25	323	32.33	96.89	4.02
	40	323	32.33	94.50	4.04
	50	324	32.43	95.93	4.01
	75	326	32.63	93.54	3.98
	100	317	31.73	84.21	3.83

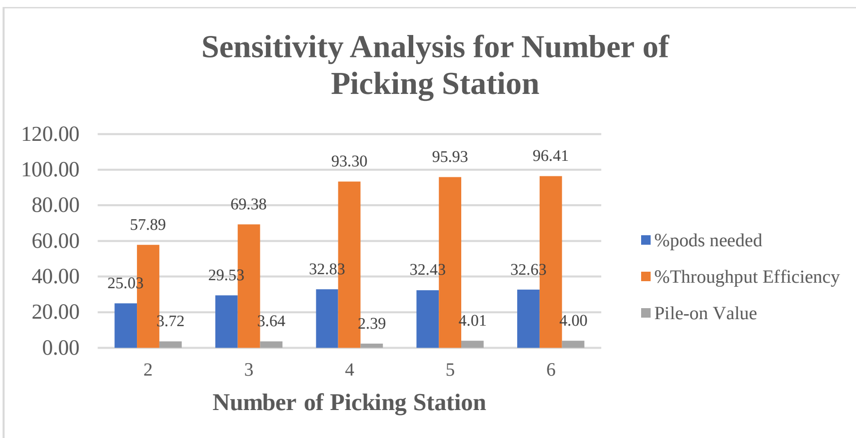


Figure 4.5 Results Comparison of Proposed Model with Different Numbers of Picking Stations

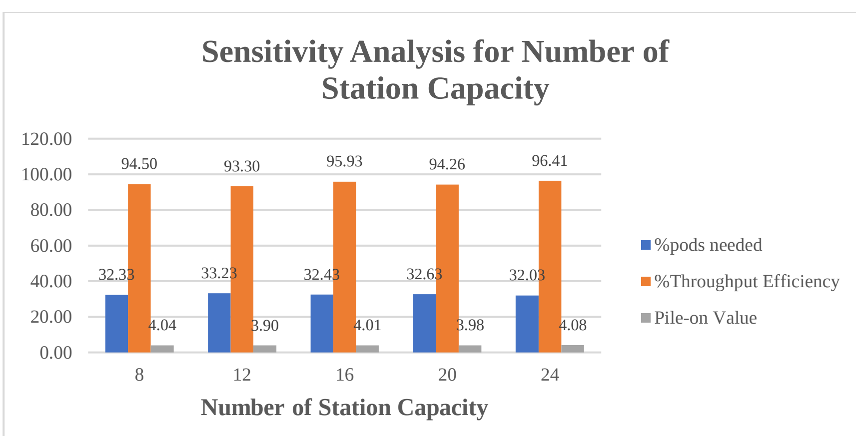


Figure 4.6 Results Comparison of Proposed Model with Different Station Capacity

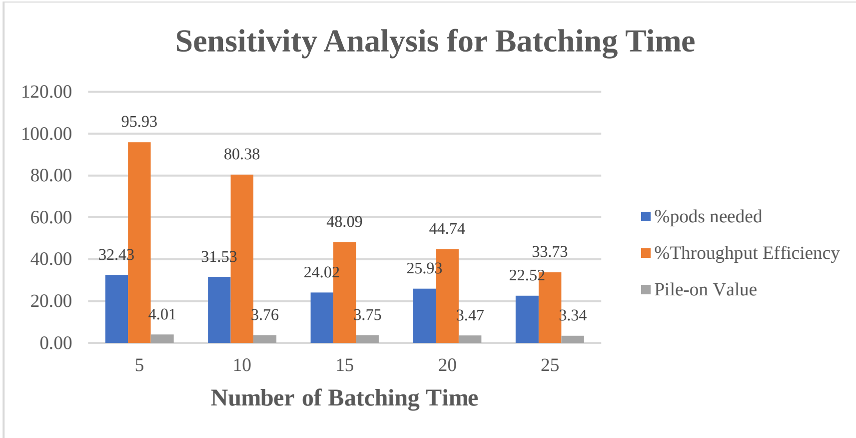


Figure 4.7 Results Comparison of Proposed Model with Different Numbers of Batching Time

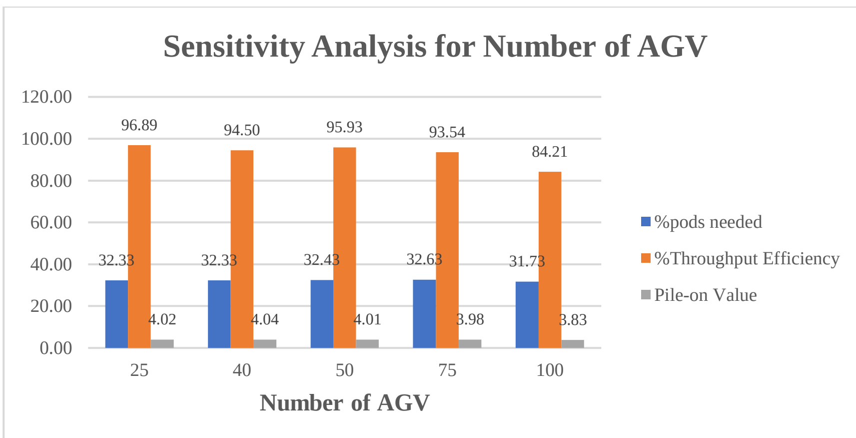


Figure 4.8 Results Comparison of Proposed Model with Different Numbers of AGV

The figure above presents the proposed model's sensitivity analysis results of different parameters.

- The Effect of Total Number of Picking Stations

  Regardless of the sensitivity analysis results, increasing the number of picking stations should increase the number of orders processed in the same setting time. Based on the result in Figure 13, increasing the number of picking stations will significantly increase the percentage of throughput efficiency. There is apparent the difference in performance affected by different picking stations. The percentage throughput significantly increases from two to four. Furthermore, between four picking stations and five picking stations, there is only a slight difference. It means that the warehouse can lower the number of picking stations at most four picking stations if they want to maintain the results of the scenario.

On the other hand, an increasing number of picking stations will not directly increase the throughput. It might only have little impact on the system, so it needs further analysis to observe how many picking stations will make more significant changes. Adding picking stations without significant improvement may lead to increased investment and operational costs but do not give impactful feedback. Furthermore, the percentage of pods needed for all numbers of picking stations is not significantly different. In terms of pile-on value, five picking stations achieve the highest pile-on. However, fluctuating pile-on value for all numbers of picking stations depends on how many orders the system can finish in one simulation time. The number of pile-on that do not have a significant difference shows that the order batching and the order assignment are impactful for maintaining effectiveness and efficiency. The number of SKUs picked from one pod is not sensitive to the number of picking stations in the system.

- The Effect of Picking Station Capacity

In terms of picking station capacity, the higher picking station capacity allows the station to process more orders concurrently. When more orders can be processed simultaneously, it will increase the possibility to finish more orders in a shorter time, which can increase the total warehouse throughput. Based on the sensitivity analysis result, the results show that changing the number of station capacity affects the warehouse's performance. There is a difference between every station capacity because of the randomness factor in the simulation. It means that the proposed model has sensitive to changes in station capacity. The throughput efficiency increases align with the increase in picking station capacity. It is proven as 24 capacity achieved the highest throughput efficiency based on the simulation results. Regarding the pile-on value, the highest pile-on is achieved by the highest station capacity. It is because more capacity means more orders, allowing a higher possibility for picking more SKUs from one pod per visit.

Adding picking station capacity can increase the performance of the warehouse. However, the magnitude of the effect obtained also depends on the parameters in the warehouse. From this research results, since the simulation hours and the number of orders is limited, increasing station capacity might not significantly impact after a specific point. With the limited number of orders that can be solved very well with a number of station capacities, increasing the capacity might not significantly affect the warehouse's performance.

- The Effect of Batching Time

  Order batching is a variable factor since there is no definite hypothesis about how long the batching time is right. It depends on the system and the behavior of the system in it. In a Robotic Mobile Fulfilment System (RMFS), order batching time can significantly impact the efficiency of order-picking activities. An RMFS typically utilizes a fleet of autonomous mobile robots to retrieve items from storage racks, allowing more precise control over the order-picking process, including the batching time. Shorter batching times can increase the efficiency of order picking in an RMFS warehouse by allowing the robots to move more quickly between storage locations and pick items in a more timely manner. This can result in faster order fulfillment and identification and retrieval of items.

  On the other hand, longer batching times can also be beneficial in allowing the system to complete a more significant batch-picking task. A longer batching time will allow the system to gather more orders before assigning them to the picking station. However, longer picking time might lead the picking station to idle for longer.

  Based on the sensitivity of the orders batching time parameter, the highest performance is achieved when batching time is five. Therefore, releasing the group of orders every 5 minutes is the most suitable decision for the proposed model.

- The Effect of the Total Number of AGV

  The difference number of AGVs in the warehouse might affect order-picking activities. More AGV in the warehouse allows higher pods to be moved simultaneously. It allows the system to undergo more tasks and finish more orders. However, higher AGV also has drawbacks because it may cause traffic congestion and affect the system's performance. Depending on the sensitivity result, the throughput efficiency, pile-on value, and percentage number of pods needed have a slight difference. The result differs when the system uses 100 AGV as the parameter. The throughput efficiency is the lowest compared with the other numbers of AGVs. For the other number of AGVs, the performance slightly decreases for the number AGVs from 50. It means that the number of AGVs will increase the system's performance until a specific point, depending on several factors. Some factors related to the optimal number of AGVs needed are the size of the warehouse, the total number of pods, and the total number of orders. At one specific point, the number of AGVs will reach optimal, so increasing the number of AGVs will not have a positive impact.

On the other hand, adding more AGVs over the optimal point will affect the congestion in the warehouse. As a result, it will decrease the performance of the system overall. More AGVs also mean the system will need more time to do the assignment and increase the computational time. Based on this situation, adding several robots not always has a positive impact. The management should consider adding AGVs in the warehouse since higher AGVs also lead to high investment, operational, and maintenance costs. The number of AGVs also needs to be adjusted with the warehouse setting, order data, and other parameters.

CHAPTER 5
CONCLUSION & FUTURE WORK

5.1 Conclusion

Order picking is a critical operation that significantly impacts the overall efficiency and productivity of the warehouse. Implementing a picker-to-parts system named RMFS replaces the manual picking method since it can increase the accuracy and productivity of the warehouse. However, managing the order-picking assignment is one of the most challenging problems of implementing RMFS for the order-picking warehouse. POA is the decision problem related to assigning orders to the picking station.

Focusing on the POA problem, this research proposed a simulation study that combines optimization with a simulation approach between order batching and pick-order assignment. The simulation will be conducted using two different scenarios. The baseline scenario did not consider the grouping order rule in the order batching activity. Then, the order assignment will follow random assignment to the picking station, with the order received first as a priority for assignment. On the other hand, the proposed model will be picked using similarity-based order batching rule to improve the system performance. The order grouping rule uses the Jaccard index as a similarity measurement for assigning the most similar order in the same batch.

The behaviors of the system are represented in the mathematical model. However, simulation is used because the simulation can capture more dynamic environments and easily compare different scenarios or behavior changes. The simulation is conducted using specific parameter settings in the NetLogo platform. Simulation is set for 8 hours with 30 replications for each scenario. The results show that the proposed model outperforms compared with the baseline model. The similarity grouping and order assignment rule proposed in this research are competitive enough to capture the real-warehouse system with its dynamic environment. Based on the statistical test, it is 95% proven that there is a significant difference between the baseline and proposed model for all evaluation criteria.

The simulation results also showed that the proposed model outperforms compare with the baseline model. It can reduce the pods needed for fulfilling orders by around 40%. Also, it can help increase the throughput by around 4% by maximizing the pile-on to almost 60% higher than the baseline results. This research conducts sensitivity analysis for four simulation parameters to analyze the effect of changes in simulation parameters. The sensitivity results show that the proposed model is sensitive to parameter changes with different impacts depending on the parameter. Based on the sensitivity analysis results, increasing the number of picking stations will increase the performance significantly only until a specific number of picking stations is. After a specific point, adding more picking stations might not significantly impact the cost of adding more picking stations.

Furthermore, increasing the station capacity will positively influence the performances since higher station capacities allow more orders to be processed concurrently. Besides, the number of batching times is sensitive to the system's performance. Since there is no exact way to decide the exact batching time needed, this sensitivity analysis can visualize the impact of different batching times. The system will achieve higher performances in optimal batching time. The performances in higher or lesser numbers of optimal will give different performances. Last, the number of AGVs will increase the performances until some point. Adding more of them may not affect the performance significantly since the system has already reached convergence at a specific point. The parameter change needs to follow sensitivity rules to maintain the current state of performance. However, changing parameters also need to consider other factors, including investment, operational, and maintenance costs. The management needs to make sure that they will get a significant impact when they change the parameter.

5.2 Future Research

This study still has many limitations that can be improved in future research. The proposed model groups orders into SKUs based on the similarity between SKUs inside the orders. Further research can consider batching the orders and considering the SKUs in pod affinity. Besides, the simulation can be run in a longer simulation time and try to use an actual data warehouse for validating the model performance. Other evaluation criteria may be added, such as the number of tasks or service level criteria, rather than only considering the higher pile-on since the company also needs to maintain customer satisfaction to compete with the other company. Furthermore, the following study can use a mathematical model and solve the optimization or use metaheuristics to sequence the orders and pods to minimize the number of tasks that need to be done.

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